



THÈSE DE DOCTORAT
DE L'UNIVERSITÉ PSL

Préparée à MINES Paris

**Advanced Hemodynamic Modelling and Simulation of Flow
Diverter in the Treatment of Intracranial Aneurysms**

Soutenue par

Pablo JEKEN RICO

Le Date de soutenance

École doctorale n°364

**Sciences Fondamentales et
Appliquées**

Spécialité

**Mathématiques Numériques,
Calcul Intensif et Données**

Composition du jury :

C. Alberto FIGUEROA Ph.D., University of Michigan	<i>Président</i>
Marek BEHR Ph.D., RWTH Aachen University	<i>Rapporteur</i>
Umberto MORBIDUCCI Ph.D., Politecnico di Torino	<i>Rapporteur</i>
Anne-Virginie SALSAC Ph.D., CNRS, Sorbonne Université	<i>Rapporteur</i>
Thomas LIEBIG Ph.D., M.D., LMU Munich	<i>Examineur</i>
Aurélien LARCHER Ph.D., Mines Paris	<i>Co-encadrant de thèse</i>
Bruno Figliuzzi Ph.D., Mines Paris	<i>Co-encadrant de thèse</i>
Elie HACHEM Ph.D., Mines Paris	<i>Directeur de thèse</i>

an meine Familie und ganz besonders an meine Oma Hilde

Contents

Contents	i
List of Figures	iii
List of Tables	v
1 Brain Arterial Hemodynamics	1
1.1 The Arterial System	3
1.1.1 Circle of Willis	4
1.1.2 Medical Imaging	6
1.1.3 Vessel Segmentation	8
1.2 Hemodynamics Simulations	9
1.2.1 Mesh Generation	9
1.2.2 Convergence Analysis	12
1.2.3 Risk Indicators	17
1.2.4 Inlet Boundary Conditions	19
1.2.5 Outlet Boundary Conditions	22
1.3 Impact of the Domain Extension	25
1.3.1 Patient-specific Artery Networks	26
1.3.2 Methods	28
1.3.3 Results	28

Contents

1.3.4 Discussion	34
Bibliography	37

List of Figures

1.1	Schematic view of a complete CoW and the ophthalmic arteries[1]. . .	5
1.2	Examples of imaging modalities	7
1.3	Sketch of the interpolation between the vessel cross-section (in the back) and the circular profile (front).	10
1.4	Projection of the cut onto a 2D plane in the reference coordinate system ξ, η	10
1.5	Extruded outlet with a length of six times its diameter.	11
1.6	Outcome of remeshing the structured extrusion.	11
1.7	Velocity magnitude along the designated line at systole for different mesh resolutions in the vessels.	14
1.8	Convergence aneurysm refinement	15
1.9	Visualization of the CoW of patient A with highlighted mesh features (orange circles). Linear isotropic refinement (left) and structured tetrahedral boundary layers (right).	16
1.10	Velocity magnitude along designated green at systole for different time step sizes.	17
1.11	WSS derivation.	18
1.12	Generalized Inflow Waveforms	21
1.13	Left: DSA image of the aneurysm of patient B. Right: Capture of the aneurysm during surgery. The red pouch is the pronounced lateral daughter sack showing notable wall thinning and immediate rupture risk.	26

1.14	Overview of cases A-C (top to bottom) in the different extensions <i>Simplified-Full</i> (left to right). <i>Simplified</i> - Neglection of the adjacent PCom. <i>Middle</i> - Simulation of ICA and proximal vessels. <i>Full</i> - Full computation of the CoW.	27
1.15	Volumetric flow through the left ICA outlets of patient A and the targeted flow by Murray's Law.	29
1.16	Maximum velocity projection of patients A-C (top to bottom) for all complexity levels <i>Simplified-Full</i> (left to right) at diastole.	30
1.17	Cross-section cuts at the IA neck of patients A-C (top to bottom) with inscribed velocity magnitudes at diastole. Velocity streamlines are superposed with the aneurysm necks to highlight the structure of the flow.	32
1.18	WSS distributions on the aneurysms A-C (top to bottom) for the different complexity levels at diastole (a) and peak systole (b).	33
1.19	OSI distributions for cases A-C (top to bottom) under different geometry extensions.	34

List of Tables

1.1	Hemodynamic parameters for various anatomical regions from [4] and measurements of [5].	4
1.2	Element count in million elements and computation time per cardiac cycle of different meshes.	29
1.3	Outflow per cardiac cycle with respect to the in total inflow of each patient. In the <i>Large</i> configuration of case C, the flow in the right MCA was the consequence of imposing Murray’s law on the vessels of the M1-M2 bifurcation. The massflow for the <i>Middle</i> and <i>Small</i> is taken at the A1 ACA segment due to the inherent cut of the domain.	31

Chapter 1

Brain Arterial Hemodynamics

Contents

1.1	The Arterial System	3
1.1.1	Circle of Willis	4
1.1.2	Medical Imaging	6
1.1.3	Vessel Segmentation	8
1.2	Hemodynamics Simulations	9
1.2.1	Mesh Generation	9
1.2.2	Convergence Analysis	12
1.2.3	Risk Indicators	17
1.2.4	Inlet Boundary Conditions	19
1.2.5	Outlet Boundary Conditions	22
1.3	Impact of the Domain Extension	25
1.3.1	Patient-specific Artery Networks	26
1.3.2	Methods	28
1.3.3	Results	28
1.3.4	Discussion	34
	Bibliography	37

Abstract

After reviewing the fluid physics and numerics, the computational framework will now be adapted to the specific conditions of the brain arterial flow. For this purpose, we consider first an anatomical description of the arterial network and its functions. Then angiographic imaging and the segmentation procedure are described, highlighting their relevance for subsequent simulations. Commonly used boundary conditions are then discussed in relationship with the extent of the simulated domain. This latter question has been the seed for the first publication that has been carried out during the course of this work “Evaluating the Impact of Domain Boundaries on Hemodynamics in Intracranial Aneurysms within the Circle of Willis”[1]. From the analysis, we show that in the three patient cases, the hemodynamics do change considerably and a thorough check of the arterial system should be carried out before simulating. To that purpose, dynamic CT imaging, clamp tests and MR images can provide the necessary information about hypoplasia, flow directions and vessel patency.

Resumé

Suite à l'étude de la physique des fluides et des méthodes numériques, nous ajusterons désormais notre approche aux spécificités de l'écoulement sanguin cérébral. Ainsi est présenté l'anatomie du réseau artériel et ses fonctions. Les méthodes d'imagerie angiographique et le processus de segmentation seront ensuite décrits, mettant en lumière leur importance pour les simulations ultérieures. Nous examinerons ensuite le maillage, en abordant les aspects de l'extrusion et de la génération des couches limites. La discussion porte ensuite sur les conditions aux limites couramment utilisées et leur pertinence pour le domaine simulé. Ce sujet a été exploré dans notre première publication, “Evaluating the Impact of Domain Boundaries on Hemodynamics in Intracranial Aneurysms within the Circle of Willis” (Évaluation de l'impact des limites du domaine sur l'hémodynamique des anévrismes intracrâniens dans le cercle de Willis) [1]. Nos résultats indiquent que l'hémodynamique peut varier de manière significative d'un patient à l'autre, ce qui souligne l'importance de l'étude du système artériel avant de procéder à des simulations. Pour ce faire, un examen de tomодensitométrie, aussi appelé CT-Scan, un test de clamp ou un IRM, peuvent permettre de mesurer directement les données nécessaires du système artériel comme l'hipoplasie, la direction des flux et la perméabilité des vaisseaux.

- ¹ To portray a physiologically accurate representation of the flow in intracranial
² aneurysms, it is crucial to understand the specific properties of brain arterial flow.

Hemodynamics in the human body may change considerably by location and resolving the flow in each of those cases may encompass additional efforts or permit simplifications. For instance, one may consider turbulence modelling and vessel compliance when dealing with the aorta. Capillaries on the other hand may require more complex rheology modelling, but allow for a simple constitutive model for the fluid motion. For this reason, we start by describing the arterial system with a dedicated emphasis on the brain vessels that are prone to aneurysms.

1.1 The Arterial System

The cardiovascular system is a complex network responsible for the circulation of blood, delivering oxygen and nutrients to tissues, and removing waste products. The aorta, the largest artery in the body, originates from the left ventricle of the heart. The blood that reaches the brain arterial system leaves the aorta through the left common carotid artery and the brachiocephalic¹ trunk, a vessel from where the left common carotid artery originates. These ascend, splitting at the height of the neck into the external and Internal Carotid Arteries (ICA). The former supplies most extracranial tissues, while the latter leads up to the base of the brain. The ICAs are the brain's principal supplier, carrying approximately 72% of the total flux[2]. The remainder is carried through the vertebral arteries, which run along the vertebral spine and merge into the Basilar Artery (BA). The flow from the BA usually joins the ICAs through the Posterior Communicating arteries (PComs) forming a loop, or more scientifically expressed, an anastomotic² network known as the Circle of Willis (CoW). The Posterior Cerebral Arteries (PCA) start from the BA and transport blood into the back side of the head, including the cerebellum.

Arteries are uniquely structured to manage high-pressure blood flow. Elastic arteries, including the aorta and its major branches, contain a high density of elastic fibres, allowing them to stretch and recoil, buffering the pressure changes from the heart's pumping action. Muscular arteries, such as the ones found in the brain, trade in the external elastic lamina³ in favour of more developed muscular layers. These regulate blood flow by vasoconstriction and vasodilation and adjust the distribution of blood to various regions [3]. This ability is particularly important when compensating for elevated or reduced levels of pressure, which can for instance originate from physical exercise or sleep. From the arteries, blood sweeps into the brain arterioles, the smallest arterial branches before arriving at capillaries. Contrary to the many other locations in the body, these lack precapillary sphincters. Elsewhere,

¹Cephalic: In or relating to the head.

²Anastomosis: A cross-connection between adjacent channels, tubes, fibres, or other parts of a network.

³Lamina: In the medical context, a thin membrane or layer of tissue.

the sphincters contract and expand controlling the perfusion of the capillary beds. The brain however requires a consistently high supply everywhere due to its high metabolism, making these control mechanisms redundant [3].

In physics terms, the flow for each of the vessel types can be characterized through the two non-dimensional Reynolds and Womersley numbers. The Reynolds number, as already introduced in the previous section, expresses the ratio of inertial versus viscous forces in a flow. Large arteries particularly the pulmonary arteries have high inertial flow, as can be seen on Table 1.1. Farther downstream, the flow becomes increasingly dominated by viscous forces, simplifying the numerical approximations. The Womersley number shown in Eq. 1.1, depends on the angular frequency of the oscillations ω , a characteristic length D and the kinematic viscosity ν .

$$\text{Wo} = D\sqrt{\omega\nu} = \sqrt{2\pi\text{ReSt}} \quad (1.1)$$

As such, the Womersley number measures the oscillatory character of a flow with respect to its viscous dissipation. The large arteries close to the heart are therefore marked with high Wo numbers due to the pronounced pulsatility, whereas more distal vessels feature smaller values due to their smoothed flow profiles and reduced Reynolds numbers.

Anatomical region	Mean Re	Peak Re	Wo
Ascending aorta	1000	4000–9000	19
Abdominal aorta	600	2500–4500	13
Coronary arteries	250	600–800	3
Carotid arteries	450	900–1200	4
Internal Carotid arteries	350	400–550	3.2
Vertebral arteries	180	100–400	2.1
Main pulmonary artery	1600	3000–4500	16
Arterioles	1	1	0.06
Capillaries	10^{-3}	10^{-3}	0.003
Inferior Vena Cava	400	1500–2500	13

Table 1.1: Hemodynamic parameters for various anatomical regions from [4] and measurements of [5].

1.1.1 Circle of Willis

The CoW and its adjacent arteries are an important study subject since they host the bulk of intracranial aneurysms [6]. As previously stated, the CoW is a ring connecting ingoing streams of the ICAs and the BA. It is believed that this redundancy

is a helpful evolutionary feature that protects the brain from the consequences of the infarction of one of its main suppliers. The interconnected vessels of the CoW in that case permit the contralateral ICA or BA to balance out the missing flux. Other vascular diseases, such as stenoses⁴, are also alleviated by redirecting the flow and reducing strain on the heart. The ICAs, as the main suppliers, have the largest diameter among the vessels of the CoW, measuring 4.2 mm in diameter along their intermediate sections [7]. The MCAs and ACAs, as the main branches of the ICA, have a diameter of around 2.5 mm and 2 mm, respectively, depending on the parent arteries' size [3]. The BA has a proximal⁵ diameter of 3 mm that in most cases decays quickly towards the distal section as a consequence of the plentiful branches and perforators along its way.

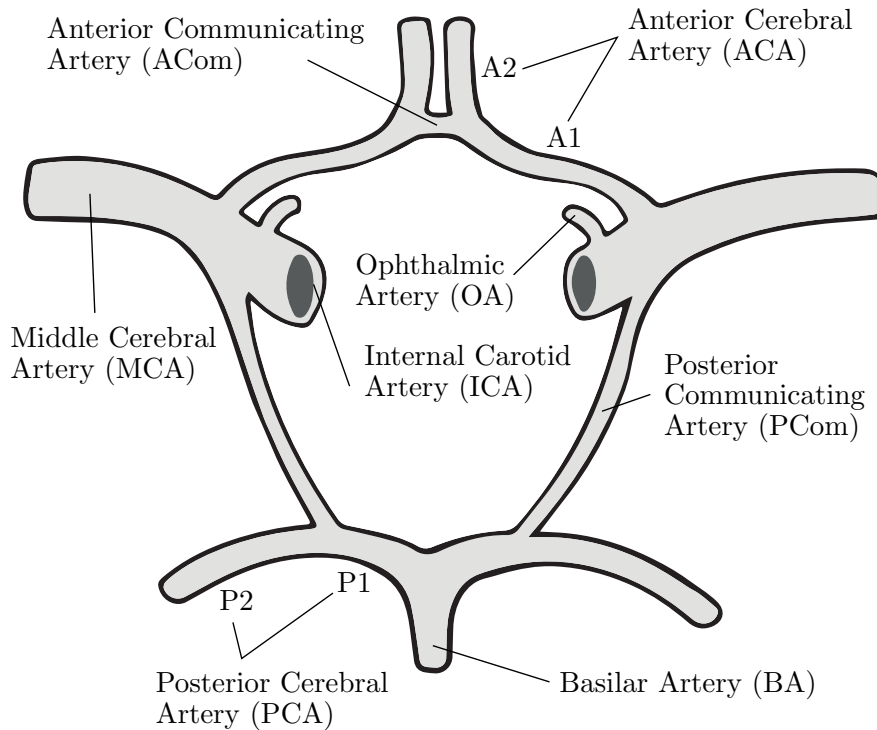


Figure 1.1: Schematic view of a complete CoW and the ophthalmic arteries[1].

The complete arrangement of the CoW, as shown in a sketch on Fig. 1.1, is not fully developed in all humans. Only an estimated 11.9% of adults possess the textbook-type CoW, while more than 50% miss at least one of the communicating vessels [8]. Hypoplastic⁶ vessels are commonly defined by having a diameter below a set thresh-

⁴Stenosis: Abnormal narrowing of a passage in the body.

⁵Proximal: Situated nearer to the body's centre or the point of attachment.

⁶Hypoplasia: Underdevelopment or incomplete development of a tissue or an organ.

old, which in the context of the CoW is often 1 mm [8, 9]. The type of CoW determines how the flow is distributed from the suppliers. One common formation is so-called fetal PComs, in which the P1 segments of the PCAs are underdeveloped and the flow to the posterior circulation is instead provided by the ICAs through the PComs [3].

The shape of the CoW has been shown to influence the likelihood of IA, doubling their prevalence on incomplete circles over complete ones [9]. The anterior circulation, comprising the ICAs, MCAs and ACAs, as well as the ACom, contains 85% of the lesions [6] and is, therefore, the most common target of operations. On it, there are three particularly prone locations, the ACom (30–35%), the PCom (30–35%) and the middle cerebral artery bifurcation (20%). Other common locations involve the ophthalmic artery, and in the case of the posterior circulation, the basilar tip and the vertebral junction [10]. Due to the asymptomatic nature of most unruptured IA, they are usually found during routine checks or unrelated screenings. The main tool for analysing the disease is angiography images obtained either from Computed Tomography (CT) or Magnetic Resonance (MR) scans. Both modalities encompass benefits and disadvantages that are also important when segmenting the patient’s 3D vessel geometry.

1.1.2 Medical Imaging

Arterial angiography⁷ highlights the structure of vessels in a designated location of the body. CT angiography involves using ionising radiation to visualise the blood vessels. A contrast dye, usually iodine-based, is administered through a catheter placed in a vein of an extremity. As the dye travels through the bloodstream, the CT scanner takes a series of detailed cross-sectional images in which the radiopaque ink and the bones stand out from the nearly transparent remaining tissue. The resulting images have a spatial resolution between 0.6 and 1 mm and are most commonly used in diagnosis [11]. Higher resolutions can be obtained with a different CT modality, the Digital Subtraction Angiography (DSA). There, the contrast agent is directly flushed into the target artery upstream of the region of interest, providing superior resolutions of 0.2–0.3 mm [11]. Additionally, a second take without a contrast agent is subtracted from the main take to remove static structures, such as bones or coils. The scan can likewise be executed by rotating the scanner during the capture. The images recorded during a rotational sweep are then reconstructed into one image cube with isotropic resolutions around 0.15–0.2 mm [11].

The other commonly employed modality is Time-Of-Flight MR angiography (TOF-MRA). TOF-MRA leverages the movement of blood to produce high-contrast images

⁷Angiography: Visualisation of vessels.

without the need for a contrast agent, which is particularly beneficial for patients with allergies or renal impairments. This method exploits the differences in magnetic properties between stationary tissue and flowing blood to create the images. In a typical MRI scan, repeated radio pulses saturate the protons in stationary tissues, making them appear darker. However, in TOF-MRA, as blood flows into the imaging slice, it hasn't been exposed to these pulses. This means that the protons in the flowing blood are not saturated and retain their signal, appearing bright in the final image. In contrast, the stationary tissue within the imaging slice has been exposed to repeated pulses and therefore appears darker. This difference in signal intensity creates a natural contrast between the bright blood vessels and the darker surrounding tissues, enabling clear and detailed visualisation of the blood vessels (see Fig. 1.2). The resolutions of TOF-MRA depend fundamentally on the intensity of the magnetic field that can be generated. Common 3 T machines yield anisotropic resolutions of $0.42 \times 0.42 \times 1.5$ mm, where the coarse side corresponds to the axis in which the scanner sweeps. Newly introduced 7 T machines require shorter scanning times and offer higher resolutions of up to 0.3 mm [12]. MR angiography, as already mentioned, profits from being a non-invasive technique that can simultaneously highlight all sections of the CoW, unlike DSA, making it a widely used diagnosis tool. During operations, clinicians use the real-time imaging capabilities of CT scanners

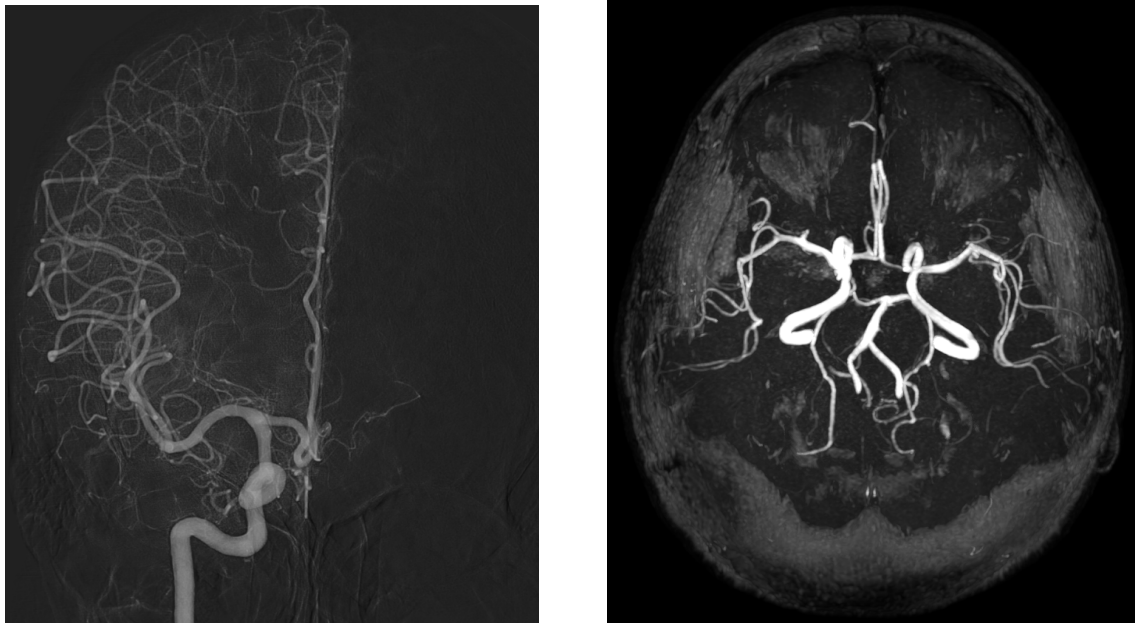


Figure 1.2: Maximum intensity projections of the brain arteries with two different imaging modalities. On the left: High-resolved DSA of one brain half. On the right: Non-invasive 3D TOF MR.

to guide catheters to the designated area. Injections of ink can then help analyse the malformations and assess the performance of the inserted prosthesis. These devices are usually made of radiopaque materials, such as platinum (coils), or contain tungsten markers to help locate them.

1.1.3 Vessel Segmentation

The images obtained during the scans are segmented by radiologists to measure possible malformations and plan interventions. In the process, attention is given to the region of interest and anatomy knowledge used to differentiate between imaging artefacts and vasculature features. When the segmentations ought to be used in hemodynamics simulations, a larger portion of the vasculature has to be processed to ensure the integrity of the simulation. As described in the introduction Berg and Voß *et al.* [13, 14] conducted a computing challenge in which they isolated the factor of the segmentation in the simulations of hemodynamics. The authors observed a sizeable dissent among some teams and recommend handling this task with care.

The 6 patient-specific cases included in this thesis have been segmented and checked with cooperating clinicians to minimize the aforementioned bias. Naturally, this does not translate into impollute work, since even clinicians assess segmentations differently. The software to carry out the segmentation is 3D Slicer, an open-source program providing handy tools to navigate and process medical image data. To obtain a smooth representation of the arterial lumen, the domain of interest is defined and extracted from the rest of the image. Subsequently, minor smoothing and gradient filtering operations can be performed to remove background noise and enhance the lumen interface. The image can then be split according to the intensity of the vessels and the background, returning a binary mask of the arteries. One common medical practice at this stage is to remove small and hypoplastic vessels from the segmentation due to the small implications and more error-prone delineation. This is commonly adopted for arteries whose diameter is below 1 mm. When the segmentation is finished, it is highly recommended to check images from another modality for confirmation. For example, one may incur errors when a portion of the vessel is situated in the plane of an MRA scan, causing its size to appear smaller. Conversely, DSA sometimes hides communicating vessels due to the flow of the contrast agent. An MRA in that case may reveal the existence of these vessels which, as will be shown later, can change notably the simulated hemodynamics.

At some point in the segmentation, it is required to limit the extension of the vessels surrounding the diseased tissue. This presents an interesting question that is not fully described in the literature, particularly concerning communicating vessels. There the flow direction is not necessarily clear *a priori*, as it might depend on the overall flows of the circuit. This interesting question has been the seed for the first

publication that has been carried out during the course of this work. Before however engaging in more depth on that question, a few more topics will be introduced that will explain key points in modelling physiological hemodynamics.

1.2 Hemodynamics Simulations

1.2.1 Mesh Generation

The segmentation yields a binary mask identifying vessels and the background. Using a resolution-dependent smoothing factor, the surface of the bodies is meshed with triangles. Hereby, the meshing algorithm provided by Slicer does not distinguish between walls and the cut inlets and outlets. Therefore, the mesh has to be manually opened with planar cuts. For carrying out this operation, Paraview and vtk can be used, which additionally permits re-triangulating the mesh to remove any quadrilateral patches originating from the cuts. The location of the inlet cut is best set on a straight segment of the parent vessel, for instance, the cervical portion of the ICA, which stretches for more than 30 mm in a straight untapered manner [15]. This alleviates the fact of modelling the inlet velocity profiles through a simplified parabolic or plug profile [16, 17]. The tortuous vessels proximal to the IA then bend and shape the flow into a physiological shape again reducing the modelling errors [18]. Additionally, it is advisable to extrude the open mesh along its open boundaries for several reasons. It further reduces inflow modelling errors and allows mapping an irregular cross-section to a circular profile. The latter is especially important when parabolic or Womersley profiles are to be used since they rely on cylindrical pipes. On the outlets' side, the extrusions reduce the risk of reflux into the systems, simplifying the CFD modelling overall.

Extrusion of inlets and outlets are generated with transfinite interpolations between the original inlet and a target circle. The target profile is generated by fitting a circle to the original cross-section and shifting it along the inlet's outward normal. The concept is sketched in Fig. 1.3. The parameterization of the extruded circle is based on the least-squares fit of a circle on the original, irregular boundary. To this end, every node and edge belonging to a boundary patch is fetched. The nodal positions of the given patch are shifted by the mean position towards the origin and registered in a matrix $X \in \mathbb{R}^{n \times 3}$, a singular value decomposition can be run on X to obtain the surface normal.

$$X = U\Sigma V^*, \quad \text{with} \quad U \in \mathbb{R}^{n \times n}, \Sigma \in \mathbb{R}^{n \times 3}, V^* \in \mathbb{R}^{3 \times 3}$$

The vector corresponding to the smallest singular value in matrix V^* provides the surface normal i.e. $\vec{n} = V_{3,j}$. It can then be used to rotate the boundary cross-section

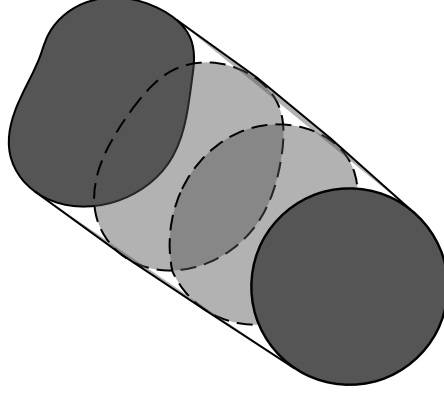


Figure 1.3: Sketch of the interpolation between the vessel cross-section (in the back) and the circular profile (front).

190 onto a 2D plane using Rodrigues' rotation:

$$M_{rot} = sI + \sqrt{1 - s^2}[\vec{k}]_{\times} + (1 - s)\vec{k} \otimes \vec{k} \quad (1.2)$$

191 Where $[\cdot]_{\times}$ indicates the skew-symmetric matrix spanned by $\cdot$ and $\vec{k}$ and s are defined
192 by:

$$\vec{k} = \vec{n} \times \vec{e}_z, \quad s = \vec{n} \cdot \vec{e}_z \quad (1.3)$$

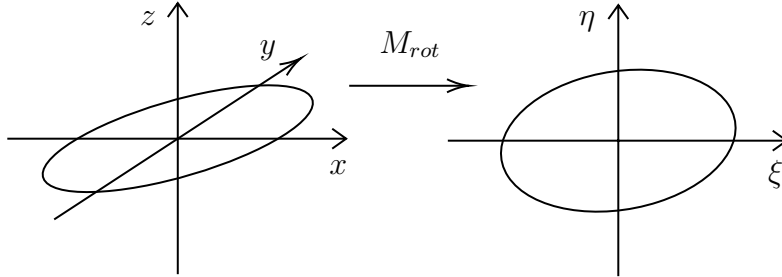


Figure 1.4: Projection of the cut onto a 2D plane in the reference coordinate system ξ, η .

Note, that all the nodes are by construction on a plane, thus making the third component of the rotated zero. The cross-section in the (ξ, η) plane can now be fitted with a simple least-squares algorithm to a circle:

$$(\xi - \xi_0)^2 + (\eta - \eta_0)^2 = r^2$$

$$[\xi_i \quad \eta_i \quad 1] \vec{c} = \xi_i^2 + \eta_i^2$$

$$\vec{c}_* = \operatorname{argmin}_{\vec{c} \in \mathbb{R}^3} \left\| A\vec{c} - \vec{b} \right\|^2$$

193 The centre coordinates ξ_0, η_0 are then rotated, translated back to the (x, y, z) space
 194 and used to parameterise a circle with the computed radius r . The circle is shifted in
 195 the outward normal direction $\vec{n}$ for a desired distance. If the cut is not perpendicular
 196 to the vessel direction, it is also possible to correct the extrusion direction by the
 197 average normal of the edges at the cross-section. The nodal coordinates are matched
 198 between the circle and the cross-section based on their minimal Euclidean distance.
 199 With a step-wise generation of nodes and a subsequent meshing, the desired mesh
 200 extension is obtained (see Fig. 1.5). The surface mesh provides a scaffold for a full
 3D volumetric mesh in which the simulations can be run. To generate a tetrahedra-

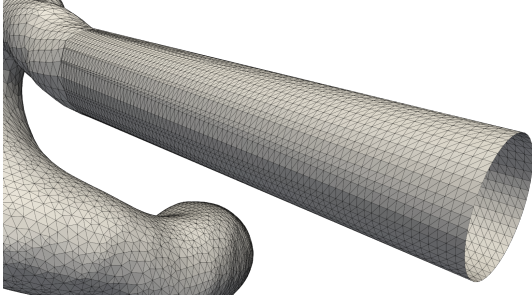


Figure 1.5: Extruded outlet with a length of six times its diameter.

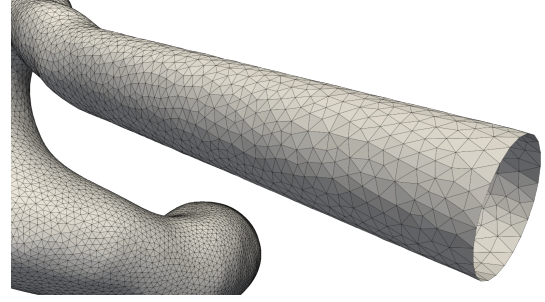


Figure 1.6: Outcome of remeshing the structured extrusion.

201 based mesh, the open-source meshing library gmsh is used. It supports the input
 202 of open *stl*⁸ files and generates high-quality meshes exploiting elaborated conformal
 203 mappings. Aside from the meshing of the interior, it is also possible to generate
 204 structured boundary layers along the vessel walls.
 205

206 **Boundary layers** play a critical role in the computation of many fluid dynamics
 207 problems. They provide a better resolution of the phenomena occurring close to
 208 walls, where velocity gradients are high and in some cases, vortices and other struc-
 209 tures can be formed and detached. In the case of hemodynamics, turbulence-related
 210 effects are not in the main spotlight. Even though studies [19, 20] have devoted
 211 their time to studying turbulence, its effects are, if present, overcast by segmenta-
 212 tion variability and boundary condition modelling errors. Instead, boundary layers
 213 play an important role in the computation of wall remodelling indicators, such as
 214 the previously mentioned Wall Shear Stress (WSS). These quantities are susceptible
 215 to layer thickness and element arrangement. In Janiga *et al.* [21], simulations were
 216 carried out using finite volume solvers on different meshes of the same geometry. In
 217 that context, using structured boundary layers resulted in a more reproducible WSS
 218 computation across meshes. This is not straightforwardly expandable towards finite

⁸STL: Stereolithography files composed of triangles in 3D.

elements, for two main reasons. Finite volumes profit from regularly aligned cells, but handle tetrahedral meshes rather poorly in comparison. Due to the intrinsic formulation of flux balances across cells, finite volumes can handle polyhedra without large changes, whereas finite elements change radically if more complex element types are considered. On the counter side, FE can increase their order of convergence by using these higher order elements while paying for a higher computation cost. While this explanation surely falls short of properly evaluating the implications on the structure of boundary layers, it is intended to show that one-to-one translations of findings between publications are to be carefully considered. Since the context of this thesis does not include higher-order elements, their influence on the boundary layer resolution could not be measured. This remains however an interesting research topic for follow-up studies. To leverage the possible bias in the WSS computation with linear elements, boundary layers of consistent size should be used.

In gmsh, the wall Boundary Layers (BLs) are generated as an inward extrusion of the surface mesh. The resulting prism elements are then split into tetrahedra. The sizes of the boundary layers can commonly be estimated by targeting a non-dimensional wall distance $y^+ < 1$:

$$h_{BL} = \frac{y^+ \mu}{\rho u_*}$$

There u_* is the friction velocity, which can be computed through the Schlichting skin-friction correlation:

$$u_* = u_\infty \frac{(2 \log_{10}(\text{Re}) - 0.65)^{-1.15}}{\sqrt{2}} \quad (1.4)$$

The Reynolds number in the model Eq. 1.4 is defined on a plate over which the flow sweeps. In our context, it remains difficult to estimate for equivalent of the length unit in such a Reynolds number. If the diameter of vessels is considered, maximal Re values span roughly between 250 and 750. In reference work [19, 20] Re numbers between 200 and 697 were estimated. Using values in this reference with overestimated inflow velocities of $u_\infty = 0.75 \text{ m/s}$, one obtains minimal boundary layer sizes of $\approx 0.05 \text{ mm}$. With the commonly adopted expansion factor 1.2, the minimal size can be progressively expanded inwards.

1.2.2 Convergence Analysis

Hemodynamics simulations intrinsically depend on the choice of resolutions in both temporal and spatial domains. To not repeat the same analysis over and over in this thesis, one illustrative case will be presented below for which both aspects have been considered. The geometry is part of the publication work that is contained in

this chapter [1]. Details about the case that are not of immediate relevance here are given in the next section section 1.3 for the sake of clarity.

Spatial Resolution

To identify appropriate choices of mesh sizes, computations can be run for different mesh resolutions. Based on the variation of primary variables or other flow indicators, it is possible to devise minimal resolution criteria. In cases of brain hemodynamics simulations, both the flow distribution through the network, as well as a high-resolution image of the flow inside the aneurysm dome are important. To properly assess the mesh convergence, the first half will be dedicated to finding a reasonable resolution in the core of the vessels and the second half will be dedicated to the bulge. By splitting the analysis, we hope to have a clearer view of the implications of mesh resolution. This has been tested on one of the cases that forms part of the study that is contained in this chapter (case C). In the first step, meshes have been generated with mesh sizes:

$$h \in \{0.1, 0.15, 0.2, 0.25, 0.3\} \text{ mm}$$

that englobe popular choices in IA literature. The aneurysm itself was kept at a fixed resolution of $h = 0.1$ mm, to study it separately. The mesh contained 3 boundary layers with a minimal size of 0.05 mm that stretched across the entire geometry. The convergence study was excuted considering the second of two cardiac cycles with a constant time step of $\Delta t = 0.001$ s. Semi-implicit schemes and non-Newtonian viscosity according to the Carreau-Yasuda law were used (see Chapter 2).

In Fig. 1.7 the velocity was plotted at the last frame of the simulation (diastole) as an example of the dynamics. It can be seen that the differences among the resolutions are overall small and of a quantitative nature, particularly, for mesh sizes below 0.2 mm. Further, we observe that the transition at $l = 4$ mm between the coarse vessel and the refined aneurysm mesh is smooth and not perceivable from the fluid plot. The flow in the aneurysm shows no alterations between the different resolutions.

The velocity fields were linearly interpolated from all meshes to the coarsest mesh for a global comparison. Taking the most refined simulation as a reference, the L_2 norm of the velocity error was computed in the whole domain and integrated over the cardiac cycle and the space. This yielded errors (in mm/s) of:

$$\varepsilon_{0.3} = 4.937, \quad \varepsilon_{0.25} = 3.559, \quad \varepsilon_{0.2} = 2.414, \quad \varepsilon_{0.15} = 1.194$$

While the average errors, confirm no large dissent among acceptable resolutions, it

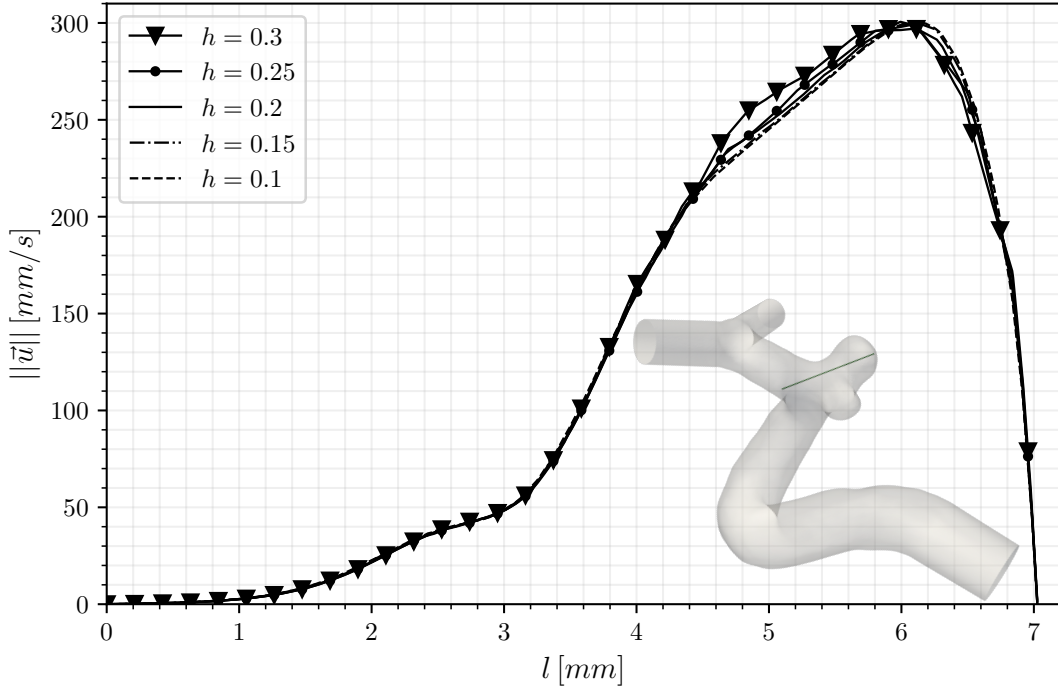


Figure 1.7: Velocity magnitude along the designated line at systole for different mesh resolutions in the vessels.

is inconclusive on the best choice among them. Instead, the previously considered comparison (see Fig. 1.7), suggest a mesh size of $h = 0.2$ mm as a good candidate.

The aneurysm refinement has been analyzed separately, refining it with isotropic mesh sizes of $h \in \{0.075, 0.1, 0.125, 0.15\}$ mm. The vessels retain $h = 0.2$ mm and all other simulation parameters are kept intact. From the obtained simulations, similar small changes in the primary variables could be observed. To consider a new facet of the important indicators, both the Oscillatory Shear Index (OSI) and the Time-Averaged Wall Shear Stress (TAWSS) have been computed.

The integrated errors with respect to the finest resolution have been recorded along with the local TAWSS error obtained from the last cardiac cycle in Fig. 1.8. The errors, in this case, have been computed by mapping the OSI and TAWSS fields from the finer meshes to the coarsest one using nearest neighbour interpolation. This interpolator, even though sub-optimal, manages to avoid a large number of undefined values which otherwise follow from the use of a smooth interpolation on meshes with a slight mismatch of the surfaces. Since the indicators (OSI and TAWSS) are restricted to the boundaries by their intrinsic nature, it is easy for nodes

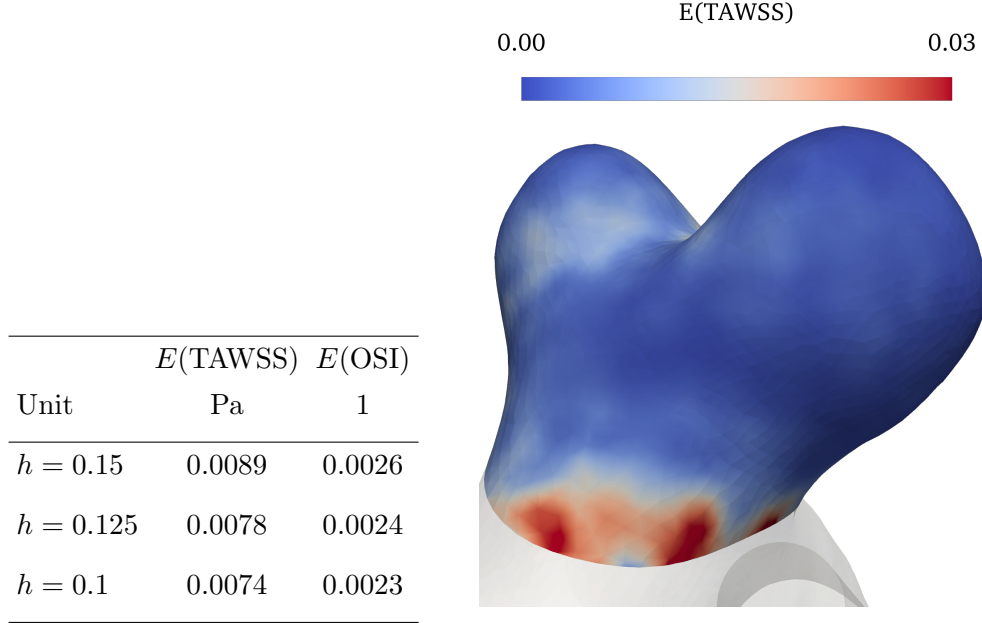


Figure 1.8: Computation of TAWSS and OSI for different levels of refinement at the aneurysm. Left: Errors for both quantities are computed by nearest neighbour interpolation from fine results onto the coarse mesh and integration over the surface. Right: TAWSS error from the coarsest resolution.

to be slightly shifted outside the intersections of the convex hulls spanned by the meshes. Any linear or higher-order interpolation fails there and yields invalid results. Nearest-neighbour interpolation however is impervious to these issues at the cost of a larger interpolation error. The obtained results show little deviations in TAWSS whose values are within the range of 0.5–2 Pa for the bulge. The maximal deviations are found on the neck sections of the aneurysm, where due to larger velocity gradients the WSS differs more notably (see right of Fig. 1.8). Similar results hold for the OSI over the bulge. The mesh size of the surface model exported from 3D Slicer lies around an estimated 0.1–0.15 mm based on the angiography image resolution. Therefore, the limiting factor from the segmentation jeopardizes the attempts to further resolve the numerical resolution. One option is smoothing and refining the segmentation surface mesh, another one is simply avoiding diminutive element sizes below 0.1 mm. From Fig. 1.8 it is concluded that a separate refinement of the aneurysm might not be necessary. Nevertheless, it will be kept at 0.1 mm, since it does not dramatically increase the computation cost. An example of a mesh using the chosen mesh sizes can be seen in Fig. 1.9.



Figure 1.9: Visualization of the CoW of patient A with highlighted mesh features (orange circles). Linear isotropic refinement (left) and structured tetrahedral boundary layers (right).

Temporal Resolution

Time step size is in general set to match a unit CFL number (introduced in Chapter 2) in the simulations. One reason for this is maintaining an equilibrium between the single components of the momentum stabilization parameter, which depends on both spatial and temporal resolutions. Having a too-small time step causes the stabilization to become homogeneous and small, *i.e.* bounded to the time step size. On one side, this minimises the added numerical diffusion. On the other, it increases the risk of losing stability. To find time step sizes that are within reasonable limits, all hemodynamics simulations of this study have been tested with different time steps $\Delta t = \{0.0001, 0.0005, 0.001, 0.0025, 0.005, 0.01\}$ s. It has been observed that for nearly all computations $\Delta t = 0.0001$ s leads to stability issues. These would also occur in some cases when using the next larger step $\Delta t = 0.0005$ s, but not for $\Delta t = 0.001$ s and above.

The time scheme also contributes to the accumulation of errors and was therefore analyzed separately with a fixed contribution in the stabilization parameters of $\Delta t = 0.001$ s. The time steps chosen are $\Delta t = \{0.0005, 0.001, 0.0025, 0.005, 0.01\}$ s, and represent common choices found in literature [22]. The velocity is plotted at systole over the same line as in the previous case in Fig. 1.10. The errors coming from the time scheme are below 0.1 mm/s for all the curves. To remain consistent with the stabilization Δt and for the sake of simplicity and computational efficiency, semi-implicit stepping will be adopted from now on with $\Delta t = 0.001$ s.

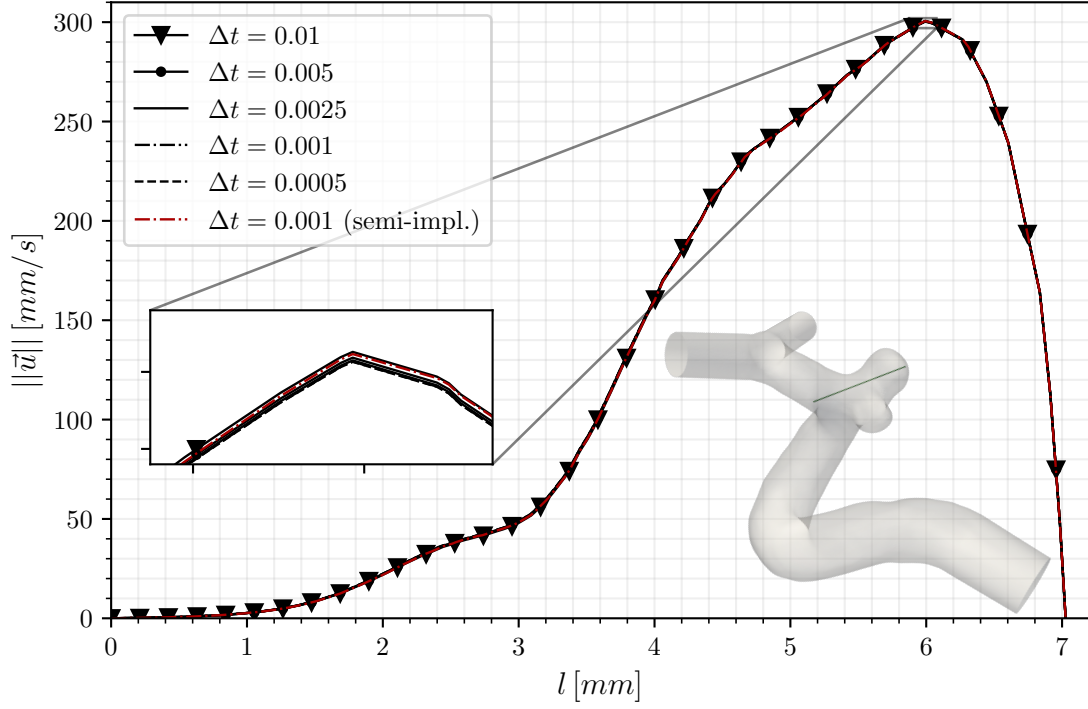


Figure 1.10: Velocity magnitude along designated green at systole for different time step sizes.

1.2.3 Risk Indicators

The objective of these carefully considered convergence studies is to ensure a proper computation of flow indicators. Most of them, as stated before, depend on the shearing exerted on the vessel walls. In its most basic representation, the WSS is expressed as:

$$\vec{\tau}_{wss} = \vec{n} \times [(\sigma \cdot \vec{n}) \times \vec{n}] \quad (1.5)$$

From an equation point of view, the WSS (τ_{wss}) is a projection of the normal stresses onto the tangential plane of the vessel. To see the visual derivation of it see Fig. 1.11. In literature, it is common to find a different, scalar definition of WSS, Eq. 1.6, which in the case of incompressibility corresponds exactly to the L_2 norm of τ_{wss} .

$$\|\vec{\tau}_{wss}\|_2 = \mu \dot{\gamma} \quad (1.6)$$

This second definition often profits from the precomputed shear rate $\dot{\gamma}$ used in non-Newtonian rheology models and does not require surface normals. When using the coarse-scale solution of the VMS stabilized Navier-Stokes, this formulation can be

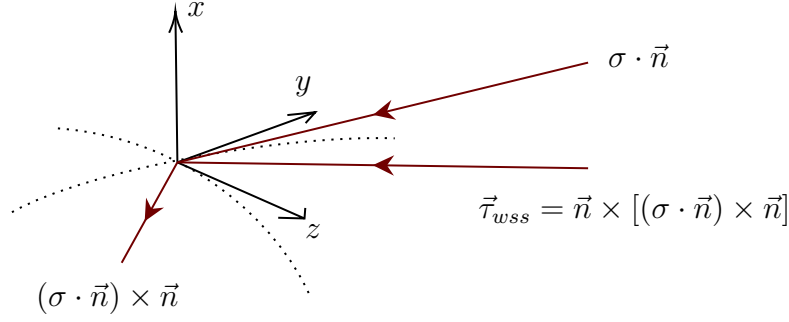


Figure 1.11: WSS derivation.

indeed problematic, yielding undesired high WSS contributions. The reason for this lies in the fact that the coarse-scale velocities are in general not divergence-free, but equal to a certain modelled subscale divergence. To provide a better picture of this, consider without loss of generality a surface patch like the one shown in Fig. 1.11, where the normal direction is $\vec{n} = \vec{e}_x$. The no-slip condition at walls enforces that the derivatives in tangential directions are zero, i.e. $\partial_y \vec{u} = 0$ and $\partial_z \vec{u} = 0$. Assume, that the fluid is not divergence-free by a factor ϵ :

$$\nabla \cdot \vec{u} = \partial_x u_x = \epsilon$$

Then the original definition of WSS from Eq. 1.5 still yields the expected contribution, since it filters out $\partial_x u_x$ in the first cross-product operation

$$\|\vec{n} \times [(\sigma \cdot \vec{n}) \times \vec{n}]\|_2 = \mu \sqrt{(\partial_x u_y)^2 + (\partial_x u_z)^2},$$

whereas the shear rate definition carries the said divergence into the WSS:

$$\mu \dot{\gamma} = \mu \sqrt{\epsilon^2 + (\partial_x u_y)^2 + (\partial_x u_z)^2}$$

343 Under these circumstances, it is recommended to use the vectorial definition of the
 344 WSS in place of the commonly used shear rate-dependent expression. Apart from
 345 alleviating mass continuity errors, it provides a resourceful quantity to measure
 346 direction in shearing, which is not possible otherwise. Many relevant indicators,
 347 such as the oscillatory shear index [23], build upon vectorial wall shearing. For these
 348 reasons, the vectorial formulation is preserved. The first directly derived quantity
 349 of interest is the Time-Averaged WSS (TAWSS), expressing how large the exerted
 350 WSS is per cardiac cycle (see Eq. 1.7).

$$\text{TAWSS} = \frac{1}{T} \int_T \|\vec{\tau}_{wss}\|_2 dt \quad (1.7)$$

351 The mentioned OSI quantifies the oscillating character of the WSS, attaining high
352 values where the WSS inverts its direction and low ones where stays unidirectional.
353 Its computation is shown in Eq. 1.8.

$$\text{OSI} = \frac{1}{2} \left[1 - \frac{\left\| \int_T \vec{\tau}_{wss} dt \right\|_2}{\int_T \left\| \vec{\tau}_{wss} \right\|_2 dt} \right] \quad (1.8)$$

354 Another family of indicators has been developed around the stability of WSS [24,
355 25], *i.e.* the structure around points of vanishing WSS, as well as contracting and
356 expanding manifolds. The latter analysis, related to the study of dynamical systems,
357 identifies whether a vessel patch is stretched, compressed or twisted by shearing
358 forces. While the stability of WSS has not been yet assessed in the same depth as
359 other remodelling indicators related to IA, it may play a role in cases like the giant
360 aneurysm of Chapter 5, where flow stability could play a decisive role.

361 A correct estimation of flow indicators depends on the computation aspects, but
362 also on the physiological compatibility between the simulated case and the real
363 phenomena occurring in the human body. One keystone is imposing Boundary
364 Conditions (BCs) that resemble reasonably well, the flow at the segments delimiting
365 the computation domain. These belong to either inlets or outlets and need to be
366 considered separately, due to their differing implications and determining factors.

367 1.2.4 Inlet Boundary Conditions

368 The entering volume flow depends on patient-specific characteristics, such as age,
369 physical state and the presence of cardiovascular complications, *e.g.* hypertension.
370 In principle, it is possible to record the blood flow velocity in the larger arteries
371 leading to the brain, *i.e.* ICAs, using either Doppler-Ultrasound velocimetry or 4D
372 flow MRI.

373 **Doppler-Ultrasound velocimetry** measures the velocity of blood flow or other
374 moving structures within the body using the Doppler effect. The technique involves
375 a transducer that emits high-frequency ultrasound waves into the body. When
376 these waves encounter moving objects, like red blood cells, they are reflected to the
377 transducer with a change in frequency. This frequency shift, known as the Doppler
378 shift, is directly proportional to the velocity of the moving object. By analysing
379 the frequency changes of the reflected waves, the ultrasound system can calculate
380 and display the speed and direction of blood flow. Doppler-Ultrasound velocimetry
381 is widely used when analysing stenosis and other blockages in the upper body and
382 limbs. In the crane, however, applying it is more complicated due to the bony
383 structures in its way. Also, the signal can be perturbed by any other larger vessel
384 that lies between the target artery and the transducer.

4D Flow MRI utilises, as the name suggests, an MR scanner to retrieve estimates about the flow velocity in one or several planes. In each plane, the scanner retrieves the signal over several cardiac cycles to filter out noise. For that reason, patients need to remain still for an extended time, making it a seldom-used technique in the medical field. In studies where 4D flow is used, resolutions have been reported to be around three times more coarse than the ones obtained in TOF MRI using the same machine [12]. Due to the lower resolution, mass and momentum conservation are not assured and neither is the quality of the estimations in recirculation zones. Nevertheless, the technique provides an unbiased profile of the fluxes in the brain that doesn't rely on assumptions related to boundary conditions of blood rheology. Much work has been invested lately into the evaluation and development of 4D flow MRI using Machine Learning to enhance the estimations and reduce scanning times [26–28].

The influence of the inflow waveform Since neither of the options is generally available, studies usually fall back onto generalised waveforms obtained from large populations. Ideally, these generalised curves are given for separate groups, *e.g.* age groups, to better fit specific characteristics [29]. Several publications have devoted time to analysing the discrepancies between using generalised curves and patient-specific ones. In 2014 Jansen *et al.* [30] and Ahn *et al.* [31] conducted independent studies with partially contradicting outcomes. In the prior a small population was the basis of simulations using different inflow waveforms with the same volume found in literature. The authors found only significant alterations in the OSI magnitudes. On the other hand Jansen [30] found that in the 36 treated patient cases, for which 2D phase-contrast MR recordings were available, substantial differences were the result of different inflow waveforms. In the last and largest study by Li *et al.* [32], 148 patients were considered. Transcranial Doppler sonography recordings were taken of the patients and used to run simulations of the IA in their brain arterial system. The same simulations, as in the previous studies, were compared to ones using generalised profiles. The study concluded that even though significant differences in the flow could be witnessed, they were not sufficiently relevant to influence the rupture risk assessment. The latter study strikes a middle ground between its two predecessors and keeps a certain degree of confidence in the use of generalised data. It is also important to note that cardiac output is a quantity subject to many alterations in the body, such as the state of activity, anaemic events and environmental changes. In that sense, it is nearly impossible to strike a “perfect” result, underlining that hemodynamic simulations are an approximation of the complex and transient phenomena occurring in real life.

Considering the presented studies and the absence of direct measurements for the patients considered in this work, generalized curves are adopted. A waveform of

the ICA obtained at the LMU University Hospital was taken as a baseline since age and other information were not present. However, the MR scans revealed the whole structure of the CoW which permitted scaling the waveform according to the indications from [33]. In the cited work, 105 healthy, young individuals are examined in regards to the flow rates in the ICAs and BA considering posterior or anterior hypoplasia. This permitted generating indications for the flow split among suppliers in the absence of a complete CoW. An example for the volume flow curve alongside another age-based curve [29] is given in Fig. 1.12.

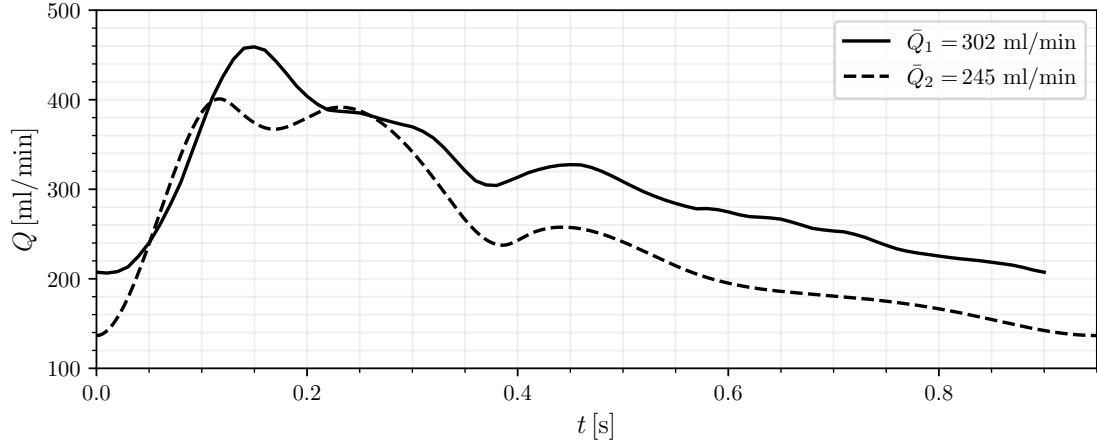


Figure 1.12: Two waveforms $\dot{Q}_1$ [33] and $\dot{Q}_2$ [29] are used in this thesis. The first is typical for adults and the second for older adults (> 60 years). The length of the cardiac cycle is $T = 0.9$ s for $\dot{Q}_1$ and $T = 0.95$ s for $\dot{Q}_2$. The shorter period contributes to the higher values of $\dot{Q}_1$, since it was scaled to a mean value of 302 ml/min [33].

431

The volumetric flow can be imposed with varying degrees of complexity. The first, and most simple is using a plug profile over the inlet. If a sufficiently long portion of vasculature, ideally containing turns [17], is used, this method may be applied [18]. Its main advantage is that it does not require a circular inlet to be implemented and can instead be applied to any surface. Another more realistic approach is the use of a parabolic profile:

$$\vec{u}_{in} = -\vec{n}_{in} \frac{2\dot{Q}}{\pi R^2} \left(1 - \frac{r^2}{R^2} \right) \quad (1.9)$$

The profile is based on a fully developed Poiseuille flow, *i.e.* for a stationary viscous flow in a circular pipe. Therefore, it needs to be applied on a circular inlet of radius R , where r is the distance to the circle's centre. Note that the normal vector $\vec{n}_{in}$ is defined outwards, like all other outflows, and thus requires to be inverted for

the profile here. Using this profile is, due to its smooth progression and simple implementation one of the most frequent choices in IA literature [34, 35]. The last approach found in common IA literature is based on the theory of oscillating blood flow described by Womersley in 1957 [36]. Here, not only the magnitude but also the inflow profile varies over time, mimicking the behaviour of a viscous fluid in a circular pipe under oscillating inflow pressures. While arguably being the most physiologically compatible flow of the presented methods, it requires some more implementation steps as well as circular profiles, much like the parabolic profile. For the present work, Womersley profiles have not been used due to a negligible impact on IA hemodynamics [37] while implicating an increase in the complexity of the simulations.

1.2.5 Outlet Boundary Conditions

Outlet boundary conditions in the brain arterial system are overall difficult to model and calibrate. Contrary to the aorta and its branches, the vessels branching off the CoW are small and measuring flow velocities in them is nearly impossible. The main difference with respect to the inflow is that, in general, there are many more outgoing vessels at different heights of the vascular tree. Therefore, it is necessary to devise a strategy to split the flow correctly throughout the sinks of the system.

Zero-pressure outlet conditions are the most common choice in IA literature [22, 38]. This effectively renders the flow distribution proportional to the resistance along the paths to each vessel. Aside from the fact that no physiological law supports this BC, it implies that the flow split becomes dependent on the length of the outlet extrusions or the outlet cut-offs, and the choice of viscosity model. This can be seen by simply modelling the flow in a network with a 0D model, much like an electrical circuit. If a ground is applied at each end, the sole factor directing the flow through the network is the resistance along each path. However, it is expected that the dominant contribution to the resistance stems actually from the smaller vascular bed of arterioles and capillaries found past the limits of our computation models. This poses a direct contradiction with the zero-pressure law, raising criticism from many [22, 34, 39, 40]. Another indirect drawback is that the overall pressures in the system are not adapted to the ones found in reality. While this does not directly affect CFD simulations, it does yield utterly wrong results when wall compliance is considered. A considerable leap in development was struck in 2006 by Vignon-Clementel *et al.* [34] who developed a reduced-order model for the outlets.

Reduced-order outlet models extend the regular 3D domain by 1D domains in which simplified versions of the Navier-Stokes equations are solved analytically. As

478 a result of the derivation, the pressure is set as a function of the current volumetric
 479 flow times a resistance. In the most simple execution, the resistance is a constant
 480 R , much like in an established Poiseuille flow.

$$p_{out} = R \int_{\Gamma_{out}} \vec{u} \cdot \vec{n} dA \quad (1.10)$$

481 If vessel compliance is considered, the resulting pressure is no longer linearly de-
 482 pendent on the volume flow but is subject to an impedance $Z(t)$. The impedance
 483 models the shift of the outgoing flow caused by the capacity of arteries to elastically
 484 deform due to pressure variations.

$$p_{out} = \frac{1}{T} \int_{t-T}^t Z(t - \tau) \int_{\Gamma_{out}} \vec{u} \cdot \vec{n} dA d\tau \quad (1.11)$$

485 The latter BC is commonly referred to as a 2-element Windkessel model. The de-
 486 nomination *Windkessel* stems from German and means wind kettle. It was
 487 used to describe water circuits with integrated air chambers that acted as a damp-
 488 ening volume absorbing shocks from the abrupt flow changes. In the context of the
 489 heart, having compliance outlet conditions is seen as a critical requirement. There,
 490 as described above, arteries absorb the shocks emitted by the heart contractions and
 491 smooth them out to ensure a continuous flow. In the brain, on the contrary, arteries
 492 lack much elasticity and instead have more developed smooth muscle cells in the me-
 493 dia [3]. This renders compliance small, which is thus difficult to observe in dynamic
 494 CT imaging. Note that the same does not necessarily apply to diseased aneurysm
 495 tissue, as is known from the literature and will be shown in the last chapter of this
 496 thesis. In summary, from the presented boundary conditions it is assumed that a
 497 linear resistance together with the volume flow rate can provide an effective way of
 498 imposing outflow pressures. The remaining question to solve is how to determine
 499 this resistance.

The Principle of Minimum Work also called Murray's Law, describes a natural law of branchings that can be exploited to estimate the split. In the course of evolution, humans have developed an optimized branching pattern in arteries that minimizes the overall maintenance P_s and transport cost P_t [41, 42]. One can think of the maintenance cost as a penalization for having just a few large vessels since blood needs to be pushed through capillaries to effectively exchange nutrition and waste products. One way of expressing the transport cost is through Poiseuille's law:

$$P_t = \dot{Q} \Delta p = \frac{8\mu l \dot{Q}^2}{\pi r^4}$$

The storage work grows proportionally to the volume of blood held by an idealised vessel, which is multiplied by a volumetric cost λ , usually set as $\lambda = 648 \text{ J}/(m^3s)$ [43].

$$P_s = \lambda V = \lambda l \pi r^2$$

To retrieve the optimal radius, the minimizer in ?? is differentiated in r . Note, that the sum of terms with a dependency r^{-4} and r^2 has one local minimum for $r > 0$.

$$\begin{aligned} \min_r P_t + P_s &= \min_r \left(\frac{8\mu l \dot{Q}^2}{\pi r^4} + \lambda l \pi r^2 \right) \\ \frac{\partial}{\partial r} (P_t + P_s) &= -\frac{32\mu l \dot{Q}^2}{\pi r^5} + 2\lambda l \pi r = 0 \\ \dot{Q} &= r^3 \sqrt{\frac{\lambda \pi^2}{16\mu}} \end{aligned}$$

For a given bifurcation with an inflow $\dot{Q}_{in}$ and several outflows $\dot{Q}_i$, Murray's law is exploited using the flux balance:

$$\begin{aligned} \frac{\dot{Q}_1}{\dot{Q}_2} &= \frac{r_1^3}{r_2^3} \\ \dot{Q}_{in} &= \sum_i \dot{Q}_i = \dot{Q}_1 + \frac{r_2^3}{r_1^3} \dot{Q}_1 + \frac{r_3^3}{r_1^3} \dot{Q}_1 + \dots \\ &= \frac{\sum_i r_i^3}{r_1^3} \dot{Q}_1 \end{aligned}$$

500 For any outlet k and a general exponent n , Murray's law fixes the average flow over
501 a cardiac cycle as:

$$\dot{Q}_k = \frac{r_k^n}{\sum_i r_i^n} \dot{Q}_{in} \quad (1.12)$$

502 Under the previously provided assumptions, Murray's law establishes a cubic pro-
503 portionality between the steady-state volumetric flow and the vessel radius. Varying
504 assumptions on flow properties or changing the dimensionality of the problem re-
505 sults in other exponents, e.g. 2D Poiseuille flow has $n = 5/2$, effects of turbulence and
506 compliance yield $n \approx 2$ [42]. Murray's law has been confirmed experimentally by
507 exhaustive *post-mortem* examinations of human brains by Helthuis *et al.* [41]. By
508 measuring the luminal radius in a large series of bifurcations, it was confirmed that
509 an exponent close to 3 rules the branching patterns. Depending on each individual
510 and the overall size of the bifurcation that is examined, the coefficient may vary
511 between $n = 2$ and $n = 3.5$. In simulated hemodynamics literature, index 3 is a com-
512 mon choice since it is based on the same assumptions as plain resistance boundary

conditions, making it a consistent choice [44]. Several publications have studied the impact of the exponent 2 and 3 for IA simulations [39, 45], confirming that $n=3$ promotes flow in larger vessels with respect to $n=2$. Nevertheless, since no measurement data was available, neither of the choices could be prioritised over the other one in this work.

Another studied aspect of the law is its locality, or in other words, what is considered one bifurcation. In Eq. 1.12 all outflows were considered at once, but one may arguably consider the mass balance individually at each bifurcation instead, as done in Chnafa *et al.* [39]. In the case of a perfectly Murray-compliant network, both approaches yield the same outcome. However, when Murray’s law is not met the outcomes vary. Consider for instance an unrealistic network, in which all vessels have the same diameter. In that case, the flow drastically increases in proximal outlets and diminishes exponentially on distal parts based on the number of bifurcations on its path.

One particular issue arising from this local Murray’s split concerns communicating vessels. These vessels link the anterior and posterior circulations, providing alternative pathways for the blood in case of ischaemia⁹ [3]. There, the local law fails in any case, since depending on which direction the bifurcations are swept through, the communicating vessels will have either one direction or another one. It is also problematic to delimit the vascular network at such a vessel since their flow direction can even be inverted between individuals [46]. Hence, Murray’s law as well as plain zero-stress boundary conditions are likely to alter results considerably in that scenario. To our surprise, even though 60–70% of all IAs at the CoW are situated at communicating arteries [11], no previous publications (to the authors’ best knowledge) have mentioned or discussed this question. Most works studying IAs at the carotid arteries do not describe the general topology of the CoW and the absence or hypoplasia of communicating segments. There are cases defining the PCom as a regular system outlet [39, 40] and others neglecting it [16, 47]. This motivates the contribution in which we attempt to evaluate and estimate the implications of using Murray’s law on patient cases with IAs at posterior communicating arteries. After having studied all the essential subjects to simulate brain hemodynamics, we are going to carry on with the study of this proposed question.

1.3 Impact of the Domain Extension

The study begins with a presentation of the patient cases and the different sizes of geometry that are generated to test the sensitivity of the boundary conditions. Then some specific models will be shortly discussed before showing the results of the

⁹Ischaemia: Inadequate blood supply to an organ or part of the body.

hemodynamics computations. The study and the chapter are subsequently finished with a discussion of the findings and a summarising conclusion.

1.3.1 Patient-specific Artery Networks

For the first exploratory work, we consider 3 patients (A-C) with unruptured saccular aneurysms at the ICA-PCoM junction. The angiography images were obtained using TOF-MRA on 1.5T and 3T MRI scanners with isotropic imaging and a 0.6 mm slice thickness. The choice of device ensures that the whole CoW is visible in the scans. The lumen was segmented under the supervision of interventional neuroradiologists at the LMU University Hospital. All aneurysm segmentations were double-checked with DSA imaging taken on the day of the respective interventions.

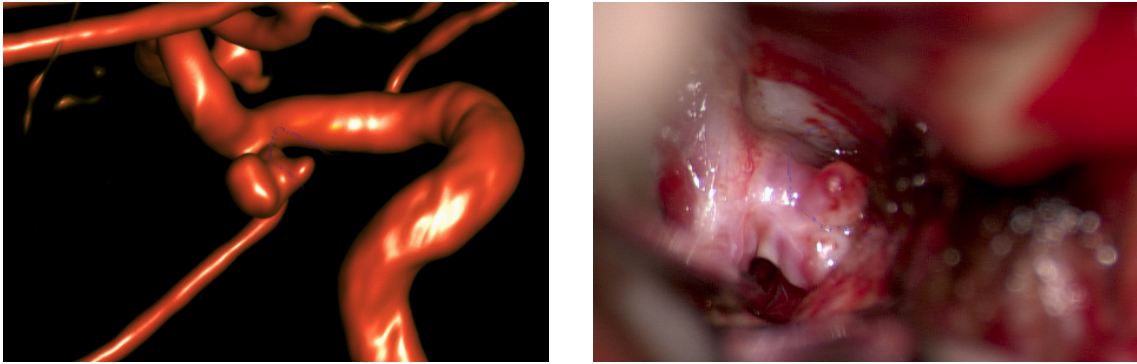


Figure 1.13: Left: DSA image of the aneurysm of patient B. Right: Capture of the aneurysm during surgery. The red pouch is the pronounced lateral daughter sack showing notable wall thinning and immediate rupture risk.

Patient A has a complete CoW with a 7.5 mm aneurysm at the left ICA-PCoM bifurcation. Patient B has a bilateral P1 hypoplasia and is therefore missing a connection of the basilar artery with both anterior circulations. The PCoMs are considered fetal since their size is larger than usual and they are the PCA's main blood supplier [3]. The aneurysm of the latter is located on the right side ($d = 5$ mm) and features a pronounced lateral daughter sack. The complex geometry of this formation was later confirmed during surgery, as seen in Fig. 1.13. Lastly, patient C too has an incomplete CoW due to an absent right PCoM. In this case, two IAs can be found, one at the left ICA-PCoM bifurcation and the other at the right MCA bifurcation. The former, being the only one considered during analysis, is classified as bilobular, due to the presence of two rounded sacks.

For each of the segmented geometries, three extensions of ascending complexity (see Fig. 1.14) are generated to assess the implications of Murray's law on simulated hemodynamics. Geometries with the keyword *Simplified* are constrained to the

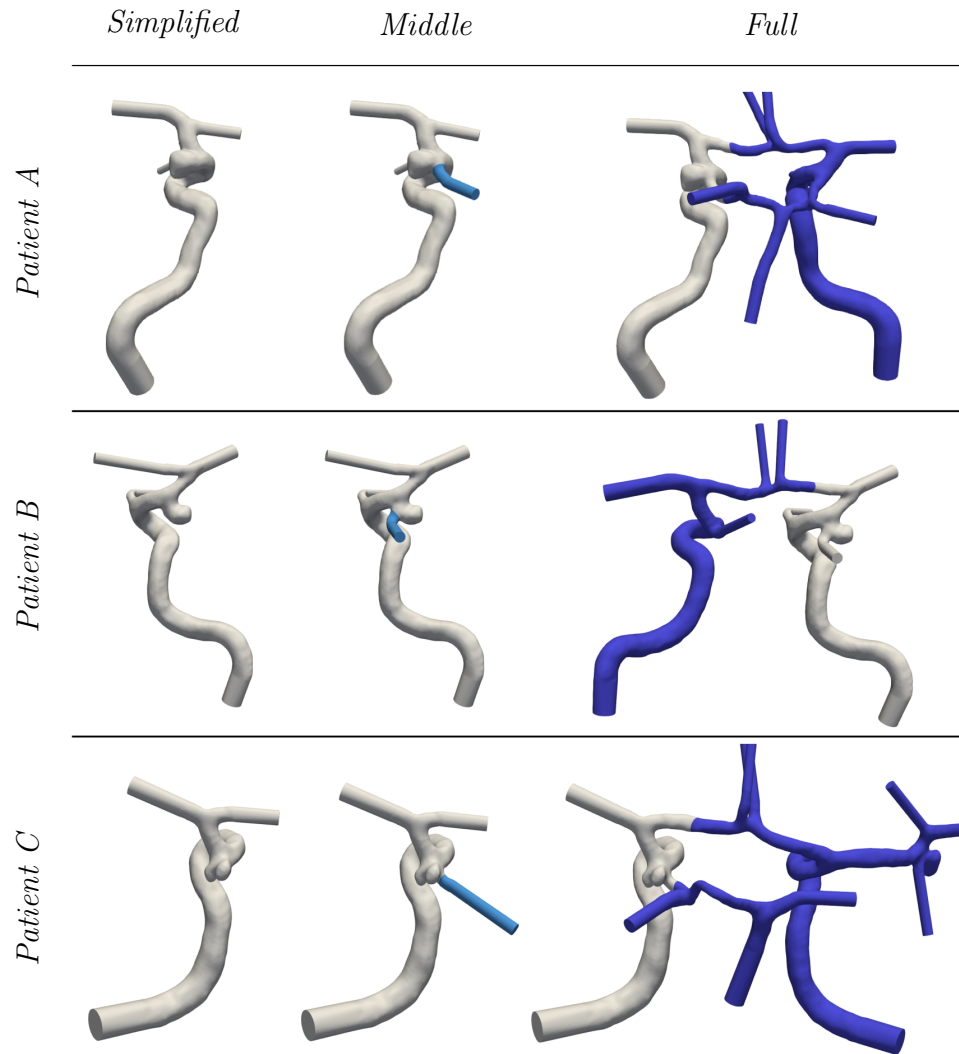


Figure 1.14: Overview of cases A-C (top to bottom) in the different extensions *Simplified-Full* (left to right). *Simplified* - Neglection of the adjacent PCom. *Middle* - Simulation of ICA and proximal vessels. *Full* - Full computation of the CoW.

main supplying artery that leads to the IA. Communicating arteries and A2 segments (see Fig. 1.1) are in this case neglected. *Middle* extends the prior *Simplified* complexity by adding the PCom. Finally, the keyword *Full* denotes the full CoW. All meshes have been generated following the steps outlined in the previous sections (see Fig. 1.9).

1.3.2 Methods

For this case study, we have used the Carreau-Yasuda model presented in Chapter 2 for all patients, since no information about the patients' hematocrit was available. Inflow conditions are described with the help of a generalized volumetric flow curve ($\dot{Q}_1$ from Fig. 1.12) which has been scaled and split among the supplying arteries. More precisely, in case A each ICA contributes 44% of the total volumetric flow while the BA carries only 12% due to its small size [33]. Cases B and C follow a split of 40%-20% between each ICA and the BA. The flow rate Q is imposed via parabolic velocity profiles at the base of the arterial system. The walls are considered fully rigid and are thus set through no-slip conditions. As previously explained, the boundary conditions are set using the linear resistance models (Eq. 1.10) and the target volumetric flow is set through Murray's law with exponent $n=3$. The resistances needed to map the targeted flow rate to pressures are computed in an iterative process. A naive Gradient Descend algorithm has been implemented to regularly adapt resistances to minimize the per-cardiac-cycle volume flow error. This procedure was executed on the 9 simulations for the study, achieving volume flows within 5% of the targeted quantity. An example of the outcome of the fitting can be seen in Fig. 1.15. To wash out the initial transient and establish a periodic solution, 3 cardiac cycles have been computed for each case due to the complexity of the flow in the *Full* simulations. The last cardiac cycle was used in the analysis of the results.

The linear system of equations was solved using a block ILU preconditioned stabilized bi-conjugate gradient method. The simulations were carried out on dual nodes (32-Core AMD EPYC 64-bit dual processor 7502, Advanced Micro Devices, Santa Clara, CA, USA) with a 2.5GHz base clock rate and an Infiniband HDR 100 interconnection. The mesh sizes and runtimes are included in Table 1.2.

1.3.3 Results

Global flow distribution

The overall flow distribution can be visually followed from the velocity magnitudes shown in Fig. 1.16. Patient A stands out due to slower flow velocities despite similar tributary flows, which is a direct consequence of having overall larger vessels com-

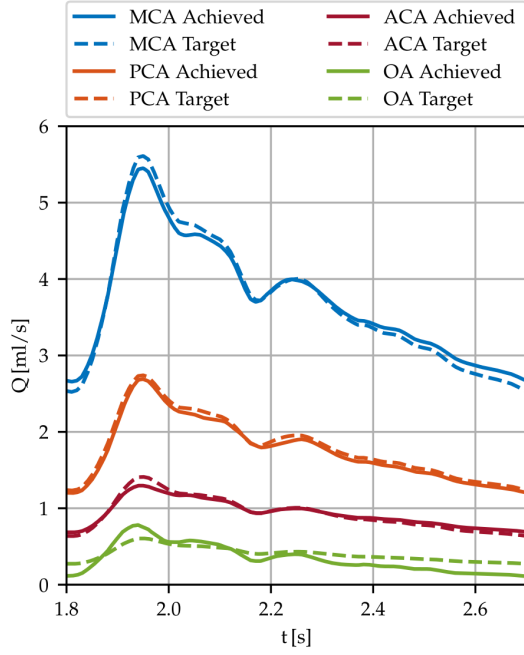


Figure 1.15: Volumetric flow through the left ICA outlets of patient A and the targeted flow by Murray's Law.

Patient	<i>Simplified</i>	<i>Middle</i>	<i>Full</i>
A	n_{elem}	3.4	3.8
	T_{comp}	9:20 h	22:10 h
B	n_{elem}	2.0	2.1
	T_{comp}	4:41 h	8:23 h
C	n_{elem}	1.6	1.7
	T_{comp}	5:07 h	8:30 h

Table 1.2: Element count in million elements and computation time per cardiac cycle of different meshes.

609 compared to B and C. The plot highlights regions of high flow speeds for instance in the
 610 terminal ICA bifurcation. The high inertia-driven flow turns mostly into the MCAs,
 611 which both due to their size and the Murray exponent ($n = 3$), carry the largest out-
 612 flow. The BA, when present, supplies the posterior circulation of the brain without
 613 directly irrigating the anterior circulation. Therefore, the flux traversing the PCom
 614 is strictly unidirectional carrying blood away from the ICA towards the PCA. Pa-
 615 tient B and C feed, according to our models, the distal PCA mostly (Patient C) or
 616 exclusively (Patient B) through the PCom due to their small or fetal P1 segments.

617 The detailed presentation of the average blood transported through each vessel is
 618 provided in Table 1.3. Notably, in the *Full* configuration, the flow through the
 619 A2 segment of the ACA is consistently smaller, exhibiting a deviation of up to
 620 5 percentage points compared to the *Middle* and *Simplified* configurations. The
 621 *Middle* configuration is characterized by guiding flow away from the MCA towards
 622 the A1 segment of the ACA, representing a distinct pattern in the distribution of
 623 blood flow. These observations are relevant for the later discussion of the biases of
 624 Murray's law.

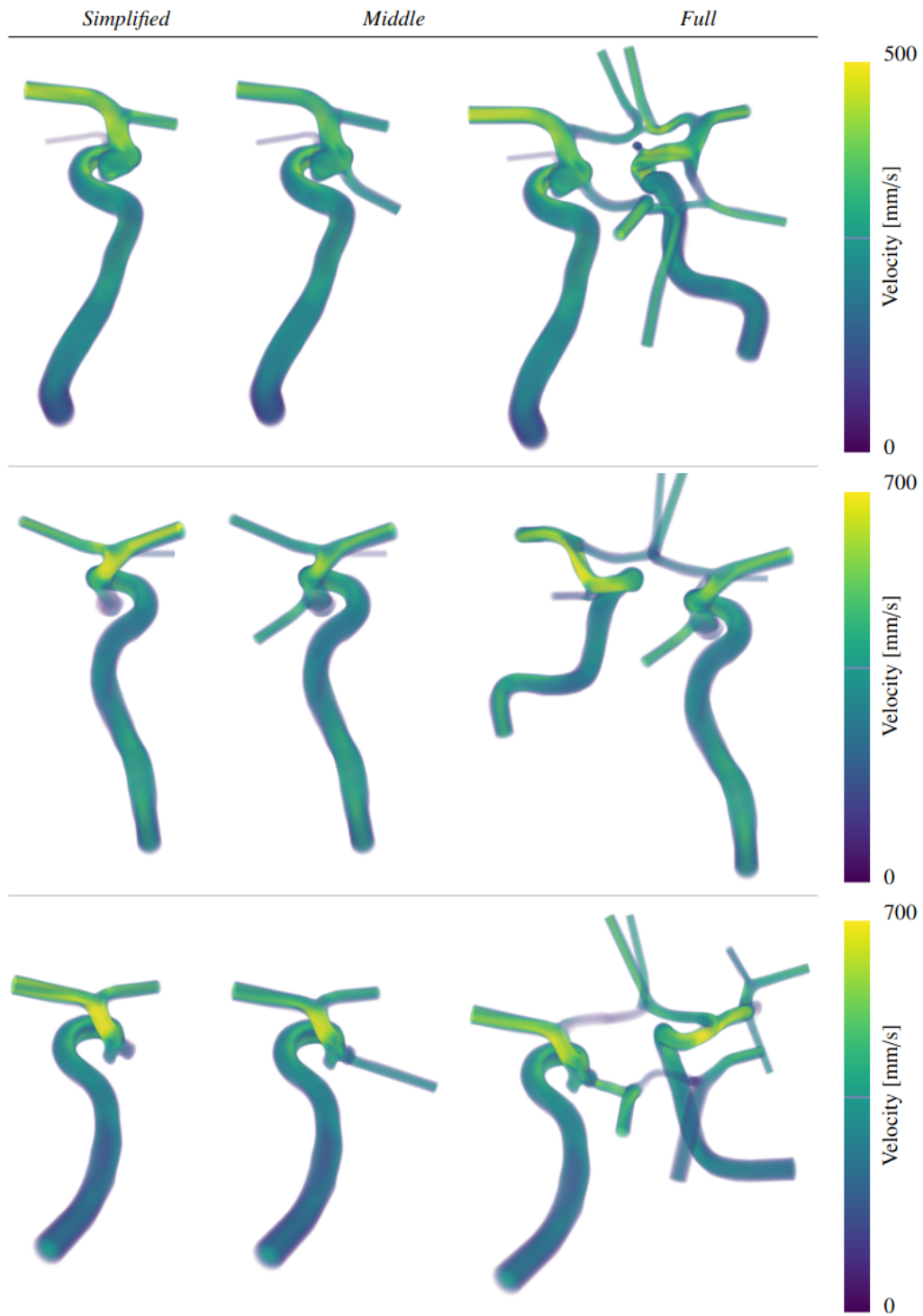


Figure 1.16: Maximum velocity projection of patients A-C (top to bottom) for all complexity levels Simplified-Full (left to right) at diastole.

Case	Complexity	MCA		PCA		ACA		OA	
		Left	Right	Left	Right	Left	Right	Left	Right
A	<i>Full</i>	28.8%	25.7%	14.0%	7.7%	7.2%	9.3%	3.1%	4.2%
	<i>Middle</i>	21.8%	-	10.6%	-	9.2%	-	2.4%	-
	<i>Simplified</i>	28.8%	-	-	-	12.1%	-	3.1%	-
B	<i>Full</i>	29.3%	21.6%	4.9%	9.6%	4.3%	7.0%	-	3.1%
	<i>Middle</i>	-	20.9%	-	9.3%	-	6.8%	-	3.1%
	<i>Simplified</i>	-	27.2%	-	-	-	8.8%	-	4.0%
C	<i>Full</i>	27.1%	35.9%	12.0%	12.4%	4.0%	8.5%	-	-
	<i>Middle</i>	22.1%	-	7.5%	-	10.3%	-	-	-
	<i>Simplified</i>	27.3%	-	-	-	12.7%	-	-	-

Table 1.3: Outflow per cardiac cycle with respect to the in total inflow of each patient. In the Large configuration of case C, the flow in the right MCA was the consequence of imposing Murray’s law on the vessels of the M1-M2 bifurcation. The massflow for the Middle and Small is taken at the A1 ACA segment due to the inherent cut of the domain.

625 Intraaneurysmal flow

626 The flow structures in the domes are visualized in Fig. 1.17 for each patient (rows)
627 and configuration (columns). The aneurysms are cut with a plane at the neck and
628 velocity streamlines of the intra-aneurysmal flow are inscribed lightly to indicate
629 the structure of the present vortices at peak diastolic time. For the inertial flow of
630 aneurysm A (first row), all three configurations show a large, centred vortex travers-
631 ing the dome that can be regarded as consistent across the three sizes. Per contrast,
632 patients B and C in the second and third rows of Fig. 1.17 display clear discrep-
633 ancies. Starting with the aneurysm B (second row) it is visible that in the simple
634 configuration, the main inflow enters the dome focused as a small jet, later decaying
635 into a stable weaker recirculation. For the *Middle* and *Full* extensions of the same
636 case, the inflow hits the neck of the daughter sack, splitting its momentum among
637 two vortices that traverse the main lobe and the daughter sack respectively. In the
638 *Full* simulation of B (second row, third column) the daughter sack’s recirculation is
639 located farther up inside the dome leading to larger in-plane velocities. Lastly, pa-
640 tient C (third row of Fig. 1.17) presents the largest disparities on both quantitative
641 and qualitative levels. One of the most noticeable differences in Fig. 1.17, is the
642 inverted circulation of the right aneurysm dome in the *Full* simulation to the other
643 two extensions. At the same time, the impinging jet’s velocities exceed the other
644 cases by, on average, 31% and 19%.

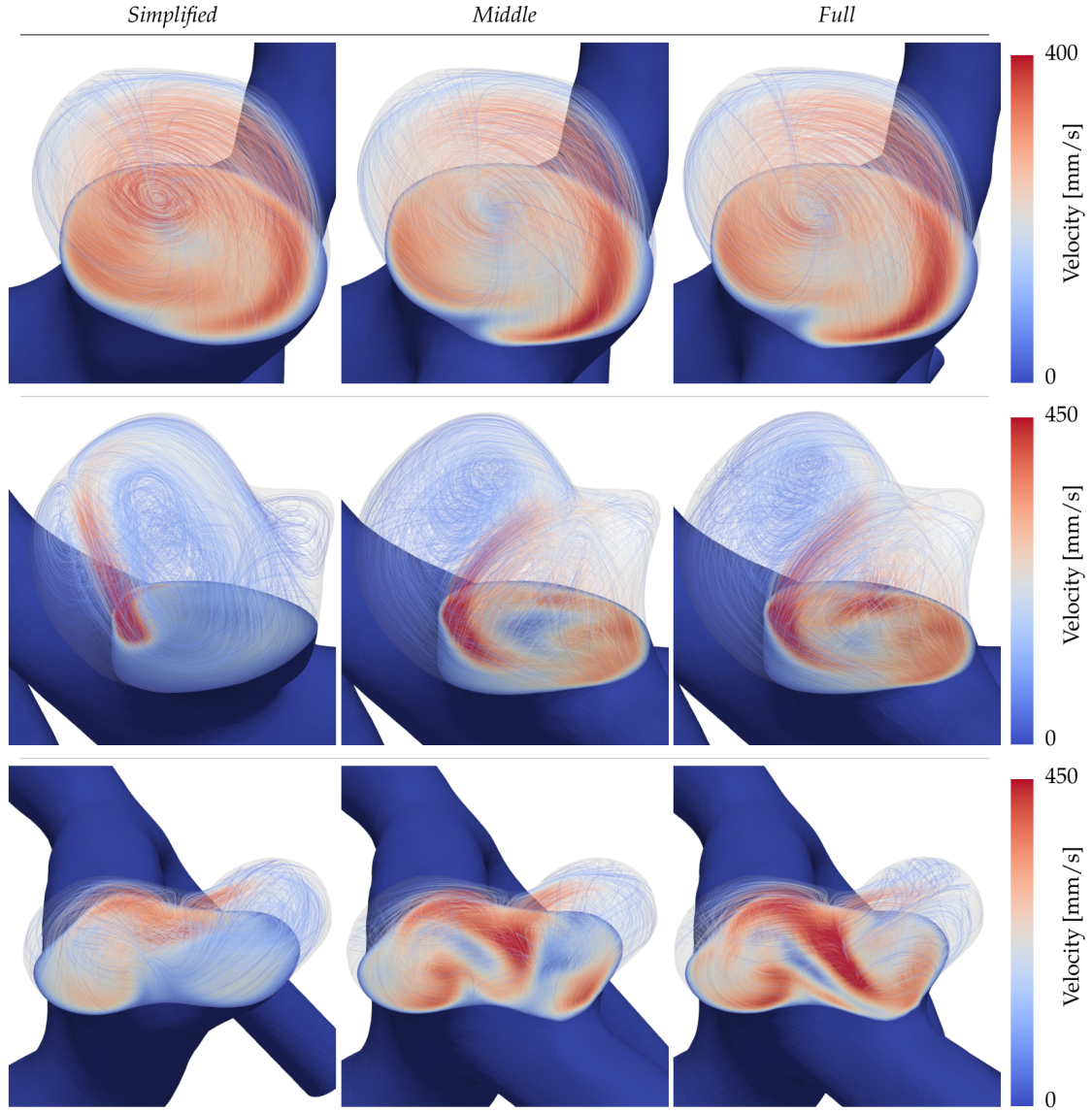


Figure 1.17: Cross-section cuts at the IA neck of patients A-C (top to bottom) with inscribed velocity magnitudes at diastole. Velocity streamlines are superposed with the aneurysm necks to highlight the structure of the flow.

645 The WSS magnitude for peak diastole and systole are shown in Fig. 1.18 (a) and
 646 (b) respectively. Patient A (first row) is characterized by a stable, qualitatively
 647 similar WSS pattern across all extensions. Here, only the *Simplified* case demon-
 648 strates smaller patches of higher shearing that are not present in the *Middle* and
 649 *Full* systems. Patients B and C instead exhibit persistently altered WSS patterns

for all of the domain extensions. Setting the *Simplified* system aside, one can observe shifts in patterns between the *Middle* and *Full* extensions in the second and third columns respectively. Most notable here is the daughter sack in patient B's aneurysm (second row of Fig. 1.18). During diastole, the WSS is relatable between the *Middle* and *Full* systems. At systole, on the contrary, the shearing increases for the *Middle* system, but remains low for the *Full* system, leading to two distinct characterizations. Following the *Full* system, the daughter sack could be identified as a risk factor due to low WSS exposure, whereas the *Middle* simulation would not. Daughter sacks play hereby an important role since their appearance is linked to an augmented risk of rupture. The last patient case in the bottom row of Fig. 1.18 (a) and (b), again shows an inconsistent WSS among the different extensions, especially aggravated at the dent separating the two sacks.

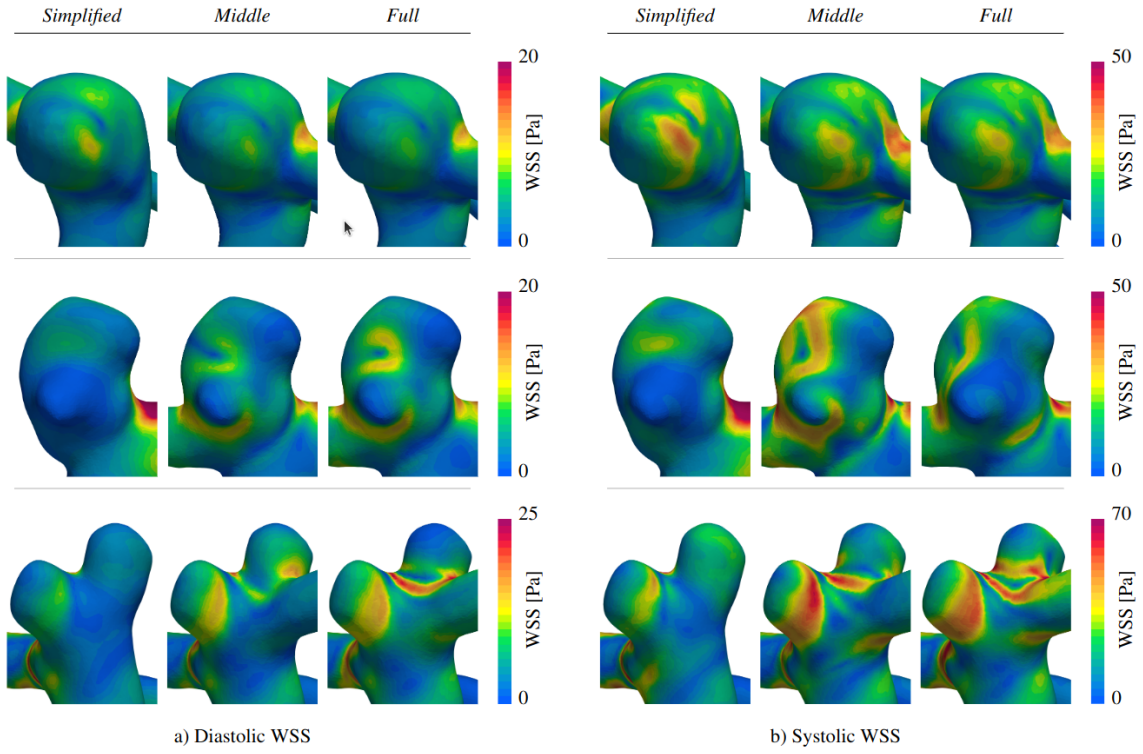


Figure 1.18: WSS distributions on the aneurysms A-C (top to bottom) for the different complexity levels at diastole (a) and peak systole (b).

661

Following the same structure of Fig. Fig. 1.18 (a) and (b), the OSI is displayed in Fig. 1.19. The observed stable WSS profile of aneurysm A inevitably leads to a nearly absent oscillatory shearing with only small localized peaks (see the first row in Fig. 1.19). Large dissent of the OSI distributions is again found in patients B and C (second and third rows of Fig. 1.19), where especially the distal lobe of patient C

666

points out. There, overall high values are found on the visible face in the *Full* CoW simulation, in contrast to the other cases.

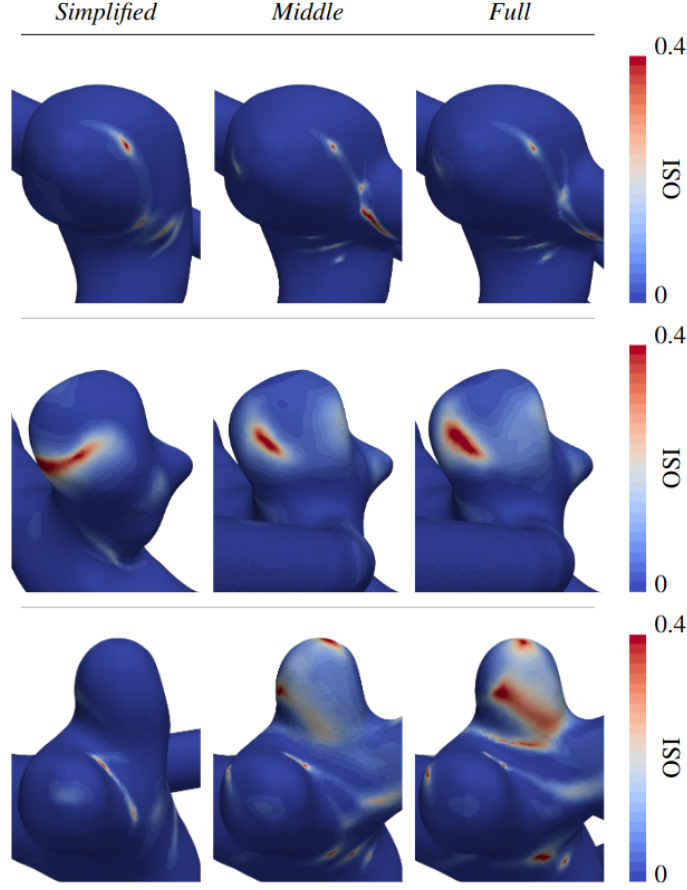


Figure 1.19: OSI distributions for cases A-C (top to bottom) under different geometry extensions.

1.3.4 Discussion

The resulting flow in the CoW resembles literature values [2] with a tendency to overestimate the flow through the MCAs. Using an exponent of ($n = 2$) in Murray's law reduces the discrepancy for the flow in the MCA at the cost of overestimating the flows in smaller vessels, such as the OA. The changes in the flow through the ACA between the different computational domain extensions are a direct consequence of Murray's law being applied on vessels of varying diameters. The proximal ACA (A1) has, in all cases, a larger radius than its post-communicating segment (A2), consequently leading to different flow rates. However, this apparent inconsistency is caused by perforator vessels around the ACom, such as the Recurrent Artery of

679 Heubner, which, due to their small size, cannot be properly simulated. Experimental
680 studies [2] support the observed decrease in flow rates. In our cases, applying
681 Murray’s law has shown a better agreement in ACA and MCA flow rates with the
682 said study when setting the domain boundary in the A1 segment.

683 The further analysis of the hemodynamics in the IA has shown differences that
684 exceed expectations, comparable to those seen in simulation challenges [13, 18, 38].
685 The *Simplified* strategy lacks physical reasoning since all of the patient geometries in
686 this study exhibit PCom arteries. Simulations confirm that neglecting these significantly
687 alters intrasaccular hemodynamics, confirming the initial intuition. Without
688 further exploration of this strategy, we conclude that it is unsuitable for the presented
689 vasculatures. The *Middle* extension, commonly found in literature, presents
690 more similarities with *Full* CoW simulations. However, moderate changes in hemodynamics
691 sometimes result in magnified alterations in WSS. These differences may
692 lead to varied correlations between flow dynamics and the aneurysm evolution [48].
693 Causes for these observations included changed boundary flow rates on one side and
694 the lack of interactions with the opposing hemisphere and the posterior circulation
695 on the other.

696 The additional blood suppliers in the case of *Full* simulations, such as the neighbouring
697 ICA and the BA, are participants in causing blood to traverse communicating
698 vessels into other circulation segments. Although most of the interhemispheric flow
699 was observed at the ACom, hence distal to the aneurysm fundus, the influence of the
700 inflow streams cannot be generally disregarded from being important [16]. In our
701 study, equal flow rates for the ICAs were imposed, aligning with a popular choice
702 shown in large-scale studies [38] due to a lack of measurement data. As stated before,
703 this is irrefutably one of the biases taken into consideration that could be lifted
704 by measurements in follow-up work. The second factor causing altered results between
705 the *Full* CoW and *Middle* simulations is, as stated, the interaction between
706 both anterior and posterior circulations. By not imposing a boundary midway in a
707 posterior communicating artery, more slack is left for the flow to unravel its natural
708 characteristics thereby providing a more physiological hemodynamic profile. Examples
709 of this, among others, are pressure drops at posterior arterial junctions and
710 bifurcations that effectively alter upstream pressure distributions and are of notable
711 relevance [40, 49].

712 The flow in these communicating vessels holds significant medical importance, given
713 that they host 60–70% of all IAs on the Circle of Willis, which itself constitutes 85%
714 of all IAs [10]. While the role of CFD in diagnosing and analyzing IA growth and
715 remodelling processes remains uncertain, it is crucial to acknowledge that modelling
716 choices significantly impact the hemodynamics of communicating vessels [46]. Moreover,
717 statistical evidence indicates that the absence or hypoplasia of certain com-

718 communicating segments is associated with a higher likelihood of developing IAs [9, 50].
719 As such, it becomes imperative to continually evaluate and refine CFD models to
720 capture the hemodynamics of the CoW more effectively. One promising outlook is
721 the use of perfusion imaging [46, 51]. Thanks to a quantification of the perfusion
722 of different brain parts, the flow into different sections of the brain can be mapped
723 through a brain atlas, giving direct feedback about the flow traversing the outlets
724 of the CoW. While this method is still in its infancy and requires an elaborate MRI
725 procedure, it holds the potential to provide a more accurate boundary condition for
726 CFD simulations.

Conclusions

In this chapter, we have presented the brain's anatomy and how to translate its complex hemodynamics into a numerical framework, including details on mesh generation and boundary conditions. The main research was dedicated to assessing the sensitivity of the aneurysm hemodynamics for varying domain extensions. To this goal, three patients harbouring ICA-PCoM aneurysms were considered in this study. For each one, simulations are run on two reduced domain extensions commonly seen in literature and contrasted with the full CoW flow. The analysis showed notable alterations of the flow in some cases biased mostly by imposing Murray's law on communicating vessels, whose primary role deviates from other arteries. The obtained results recommend taking into account the overall anatomy of each patient and considering possible implications of the flow from contralateral or posterior circulations. By confirming hypoplasia or the absence of communicating segments on MR angiography, studies can further reinforce their choice of BCs or motivate a sensitivity study otherwise.

727

Bibliography

- [1] P. Jeken-Rico, A. Goetz, P. Meliga, A. Larcher, Y. Özpeynirci, and E. Hachem, *Fluids* **9**, 1 (2023).
- [2] L. Zarrinkoob, K. Ambarki, A. Wåhlin, R. Birgander, A. Eklund, and J. Malm, *Journal of Cerebral Blood Flow and Metabolism* **35**, 648 (2015).
- [3] D. S. Liebeskind and L. R. Caplan, *Frontiers of Neurology and Neuroscience* **40**, 1 (2016).
- [4] K. Menon, Z. Hu, and A. L. Marsden, *Flow* **4** (2024).
- [5] J. R. Cebal, C. M. Putman, M. T. Alley, T. Hope, R. Bammer, and F. Calamante, *Journal of Engineering Mathematics* **64**, 367 (2009), ISSN 00220833.
- [6] R. D. Brown and J. P. Broderick, *The Lancet Neurology* **13**, 393 (2014).
- [7] R. A. Baz, C. Scheau, C. Niscoveanu, and P. Bordei, *Medicina (Lithuania)* **57** (2021).
- [8] L. B. Hindenes, A. K. Håberg, L. H. Johnsen, E. B. Mathiesen, D. Robben, and T. R. Vangberg, *PLoS ONE* **15** (2020).
- [9] L. B. Hindenes, T. Ingebrigtsen, J. G. Isaksen, A. K. Håberg, L. H. Johnsen, M. Herder, E. B. Mathiesen, and T. R. Vangberg, *Journal of the Neurological Sciences* **452** (2023).
- [10] L. Hacein-Bey and J. M. Provenzale, *American Journal of Roentgenology* **196**, 32 (2011), PMID: 21178044, <https://doi.org/10.2214/AJR.10.5329>.
- [11] L. Hacein-Bey and J. M. Provenzale, *American Journal of Roentgenology* **196**, 32 (2011).
- [12] P. Berg, D. Stucht, G. Janiga, O. Beuing, O. Speck, and D. Thévenin, *Journal of Biomechanical Engineering* **136**, 1 (2014).
- [13] P. Berg, S. Voß, S. Saalfeld, G. Janiga, A. W. Bergersen, K. Valen-Sendstad, J. Bruening, L. Goubergrits, A. Spuler, N. M. Cancelliere, et al., *Cardiovascular Engineering and Technology* **9**, 565 (2018).
- [14] S. Voß, O. Beuing, G. Janiga, and P. Berg, *PLoS ONE* **14**, 1 (2019).
- [15] M. Vijaywargiya, R. Deopujari, and S. A. Athavale, *Anatomy and Cell Biology* **50**, 163 (2017).
- [16] M. A. Castro, C. M. Putman, and J. R. Cebal, *American Journal of Neuroradiology* **27**, 1703 (2006).
- [17] S. Hodis, S. Kargar, D. F. Kallmes, and D. Dragomir-Daescu, *American Journal of Neuroradiology* **36**, 737 (2015).
- [18] P. Berg, S. Saalfeld, S. Voß, O. Beuing, and G. Janiga, *Neurosurgical Focus* **47**, 1 (2019).
- [19] K. Valen-Sendstad, K.-A. Mardal, M. Mortensen, B. A. P. Reif, and H. P. Langtangen, *Journal of Biomechanics* **44**, 2826 (2011).
- [20] K. Jain, S. Roller, and K.-A. Mardal, *Computers and Fluids* **127**, 36 (2016), 1512.02504.

Bibliography

- [21] G. Janiga, P. Berg, O. Beuing, M. Neugebauer, R. Gasteiger, B. Preim, G. Rose, M. Skalej, and D. Thevenin, *Biomedizinische Technik* **58**, 303 (2013).
- [22] P. Berg, S. Voß, G. Janiga, S. Saalfeld, A. W. Bergersen, K. Valen-Sendstad, J. Bruening, L. Goubergrits, A. Spuler, T. L. Chiu, et al., *International Journal of Computer Assisted Radiology and Surgery* **14**, 1795 (2019).
- [23] H. Meng, V. M. Tutino, J. Xiang, and A. H. Siddiqui, *High WSS or Low WSS? Complex interactions of hemodynamics with intracranial aneurysm initiation, growth, and rupture: Toward a unifying hypothesis* (2014).
- [24] U. Morbiducci, V. Mazzi, M. Domanin, G. De Nisco, C. Vergara, D. A. Steinman, and D. Gallo, *Annals of Biomedical Engineering* **48**, 2936 (2020).
- [25] V. Mazzi, D. Gallo, K. Calò, M. Najafi, M. O. Khan, G. De Nisco, D. A. Steinman, and U. Morbiducci, *Biomechanics and Modeling in Mechanobiology* **19**, 1403 (2020).
- [26] A. Harloff, T. Zech, F. Wegent, C. Strecker, C. Weiller, and M. Markl, *American Journal of Neuroradiology* **34**, 1407 (2013).
- [27] D. R. Rutkowski, A. Roldán-Alzate, and K. M. Johnson, *Scientific Reports* **11** (2021).
- [28] M. Cherry, Z. Khatir, A. Khan, and M. Bissell, *Scientific Reports* pp. 1–16 (2022).
- [29] Y. Hoi, B. A. Wasserman, Y. J. Xie, S. S. Najjar, L. Ferruci, E. G. Lakatta, G. Gerstenblith, and D. A. Steinman, *Physiol Meas* **31**, 291 (2011).
- [30] I. Jansen, J. Schneiders, W. Potters, P. Van Ooij, R. Van Den Berg, E. Van Bavel, H. Marquering, and C. B. Majoie, *American Journal of Neuroradiology* **35**, 1543 (2014).
- [31] J. Xiang, A. H. Siddiqui, and H. Meng, *Journal of Biomechanics* **47**, 3882 (2014), NIHMS150003.
- [32] W. Li, S. Wang, Z. Tian, W. Zhu, Y. Zhang, Y. Zhang, Y. Wang, K. Wang, X. Yang, and J. Liu, *Neuroradiology* **62**, 1485 (2020).
- [33] H. Tanaka, N. Fujita, T. Enoki, K. Matsumoto, Y. Watanabe, K. Murase, and H. Nakamura, *American Journal of Neuroradiology* **27**, 1770 (2006), <https://www.ajnr.org/content/27/8/1770.full.pdf>.
- [34] I. E. Vignon-Clementel, C. A. Figueroa, K. E. Jansen, and C. A. Taylor, *Computer Methods in Applied Mechanics and Engineering* **195**, 3776 (2006).
- [35] S. Lynch, N. Nama, and C. A. Figueroa, *Scientific Reports* **12**, 1 (2022).
- [36] J. R. Womersley, *Physics in Medicine & Biology* **2**, 178 (1957).
- [37] P. Berg, C. Roloff, O. Beuing, S. Voß, S. I. Sugiyama, N. Aristokleous, A. S. Anayiotos, N. Ashton, A. Revell, N. W. Bressloff, et al., *Journal of Biomechanical Engineering* **137** (2015).

- [38] K. Valen-Sendstad, A. W. Bergersen, Y. Shimogonya, L. Goubergrits, J. Bruening, J. Pallarès, S. Cito, S. Piskin, K. Pekkan, A. J. Geers, et al., *Cardiovascular Engineering and Technology* **9**, 544 (2018).
- [39] C. Chnafa, O. Brina, V. M. Pereira, and D. A. Steinman, *American Journal of Neuroradiology* **39**, 337 (2018).
- [40] C. Chnafa, K. Valen-Sendstad, O. Brina, V. M. Pereira, and D. A. Steinman, *Journal of Biomechanics* **51**, 83 (2017).
- [41] J. H. Helthuis, T. P. van Doormaal, B. Hillen, R. L. Bleys, A. A. Harteveld, J. Hendrikse, A. van der Toorn, M. Brozici, J. J. Zwanenburg, and A. van der Zwan, *Anatomical Record* **302**, 1434 (2019).
- [42] J. Keelan and J. P. Hague, *Scientific reports* **11**, 5408 (2021).
- [43] J. Keelan, E. M. Chung, and J. P. Hague, *Physics in Medicine and Biology* **64** (2019).
- [44] A. L. Marsden, J. A. Feinstein, and C. A. Taylor, *Computer Methods in Applied Mechanics and Engineering* **197**, 1890 (2008).
- [45] S. Saalfeld, S. Voß, O. Beuing, B. Preim, and P. Berg, *International Journal of Computer Assisted Radiology and Surgery* **14**, 1805 (2019).
- [46] H. Zhang, N. Fujiwara, M. Kobayashi, S. Yamada, F. Liang, S. Takagi, and M. Oshima, *Annals of Biomedical Engineering* **44**, 2351 (2016).
- [47] K. D. Dennis, D. F. Kallmes, and D. Dragomir-Daescu, *Journal of Biomechanics* (2017).
- [48] K. Valen-Sendstad and D. A. Steinman, *American Journal of Neuroradiology* **35**, 536 (2014).
- [49] J. P. Mynard and K. Valen-Sendstad, *International Journal for Numerical Methods in Biomedical Engineering* **4179**, 23 (2015).
- [50] L. Feng, H. J. Mao, D. D. Zhang, Y. C. Zhu, and F. Han, *Frontiers in Neurology* **13** (2023).
- [51] J. Schollenberger, N. H. Osborne, L. Hernandez-Garcia, and C. A. Figueroa, *Frontiers in Bioengineering and Biotechnology* **9**, 1 (2021).

RÉSUMÉ

Les anévrismes intracrâniens sont fréquents mais mal compris, malgré les conséquences graves liées à leur rupture. La complexité des processus biochimiques du cerveau et la fragilité de son environnement limitent les études in-vivo. L'étude de la dynamique des fluides numérique (CFD) peut aider à surmonter ces défis en reproduisant les conditions d'écoulement du cerveau, qui sont supposées influencer l'initiation, la croissance et la rupture des anévrismes. Les simulations CFD peuvent également permettre de tester différents traitements et évaluer leur impact. Cette thèse se concentre sur le développement d'un framework numérique pour simuler l'hémodynamique dans le cerveau, dans le but d'améliorer notre compréhension des anévrismes intracrâniens et d'optimiser leur traitement.

La thèse commence par explorer les différents aspects médicaux des anévrismes cérébraux, notamment leur forme, leur taille, ainsi que les options de traitement et les risques associés. Elle aborde ensuite la physique et les méthodes numériques qui constituent la base du framework, en apportant de nouvelles contributions méthodologiques. Fort de ces connaissances et validations, le travail se concentre ensuite sur l'adaptation des simulations à l'environnement vasculaire cérébral, en abordant des aspects essentiels tels que la segmentation, les conditions aux limites, ainsi que la définition et la portée du domaine simulé. Pour simuler le déploiement d'un stent à diversion de flux, un algorithme basé sur des contraintes géométriques est développé et rigoureusement validé à l'aide d'exemples tirés de la littérature, de données expérimentales et de cas réels de patients. Dans le dernier chapitre, ces outils sont appliqués à trois cas complexes de patients présentant des anévrismes rompus. Les effets du traitement par stent à diversion de flux sont finalement analysés et comparés à la littérature existante.

MOTS CLÉS

Anévrismes intracrâniens, hémodynamique, dynamique des fluides numérique, méthode des éléments finis, méthode variationnelle multi-échelle

ABSTRACT

Intracranial aneurysms are common but not well understood, despite the severe consequences of rupture. The brain's complex biochemical processes and delicate environment limit in-vivo studies. Computational Fluid Dynamics (CFD) can help overcome these challenges by replicating the brain's flow conditions, which are thought to influence the initiation, growth, and rupture of aneurysms. CFD simulations can also test hypothetical treatments and evaluate their impact. This thesis focuses on developing a high-fidelity computational framework to simulate brain hemodynamics, aiming to enhance our understanding of intracranial aneurysms and improve their treatment.

The thesis begins by exploring the medical aspects of cerebral aneurysms, including their various types, treatment options, and associated risks. It then delves into the physics and numerical methods that form the foundation of the simulation framework, offering new contributions to the methodology. With this knowledge and validation in hand, the work progresses to adapt the simulations to the brain's vascular environment, discussing critical aspects such as segmentation, boundary conditions, and the extent of the simulated domain. To virtually recreate the deployment of a flow diverter stent, an algorithm based on geometrical constraints is developed and thoroughly validated using literature examples, experimental data, and real patient cases. In the final chapter, these tools are applied to compute the hemodynamics in three complex patient cases involving ruptured aneurysms. The effects of flow diverter treatment are analyzed and compared with state-of-the-art medical literature.

KEYWORDS

Intracranial aneurysms, Haemodynamics, Computational Fluid Dynamics, Finite Elements, Variational Multi-scale Method